NeuroImage 318 (2025) 121349

Contents lists available at ScienceDirect

NeuroImage

journal homepage: www.elsevier.com/locate/ynimg

Individual differences in neurophysiological correlates of post-response adaptation: A model-based approach

Anna Grabowska [a,b,*], Filip Sondej [a], Julia Haaf [c,1], Michael D. Nunez [d,1], Magdalena Senderecka [a,*,1]

[a] Centre for Cognitive Science, Jagiellonian University, Krakow, Poland
[b] Doctoral School in the Social Sciences, Jagiellonian University, Krakow, Poland
[c] Psychology Department, University of Potsdam, Potsdam, Germany
[d] Psychological Methods, University of Amsterdam, Amsterdam, The Netherlands

ARTICLE INFO

Dataset link: https://github.com/abelowska/jointErrorCMD, https://osf.io/jszyh/

Keywords:
Error-related negativity
Correct-related positivity
Individual differences
Post-error adaptation
Drift diffusion model
Hierarchical Bayesian model

ABSTRACT

The cognitive control system continuously monitors actions and adapts to enhance task performance and prevent errors. More pronounced amplitudes of early neurophysiological response monitoring signals, such as error-related negativity (ERN) and correct-related negativity (CRN), have been linked to increased response caution and heightened attention to task-relevant features in subsequent trials. However, it remains unclear whether these population-level effects accurately capture adaptive processes at the individual level. This information is crucial for assessing the generalizability of findings, as effects significant at the population level may be driven by only a small subset of participants. Using Bayesian hierarchical drift diffusion modeling on behavioral and EEG data from four datasets, we found robust population-level associations between ERN amplitude and adaptive behaviors but also identified substantial individual variability. The relationship between ERN amplitude and response caution was highly context-dependent, varying across individuals, datasets, and conditions. In contrast, the association between ERN amplitude and attentional focus was more consistent, though influenced by traits related to negative affect. Our results emphasize the importance of considering individual differences to better understand the neural mechanisms underlying adaptive behavior.

1. Introduction

Successful task performance relies on an internal response monitoring system that evaluates our actions and prompts behavioral adaptations to enhance outcomes. In recent years, many studies have explored the link between response monitoring and adaptive behavior (see Danielmeier and Ullsperger, 2011; Ullsperger et al., 2014; Wessel, 2018, for a review). Most of these studies have focused on post-error behavioral adjustments, such as post-error slowing, which describes the tendency to slow down after committing an error (Rabbitt and Rodgers, 1977). Post-error slowing may be associated with adaptation (slowing to prevent future errors, e.g., Allain et al., 2009; Rabbitt, 1966; Dutilh et al., 2012; Mattes et al., 2022) or disruption (slowing due to increased arousal, frustration, or distraction of attention, e.g., Notebaert et al., 2009), but there is still debate about which explanation is more appropriate.

Post-error slowing could also be marked by large individual differences that are obscured by group-level effects. Experimental approaches often overlook individual differences or treat them as noise or measurement error. However, these differences may reflect varying cognitive strategies, decision-making processes, and individual cognitive capacities. Consequently, assuming that group-level effects apply to individuals can be misleading (Fisher et al., 2018; Haaf and Rouder, 2017). To fully understand the cognitive processes underlying adaptive behavior, it is crucial to consider between-participant variance as valuable data rather than mere noise. In the present study, we addressed this challenge by employing Bayesian hierarchical cognitive modeling on behavioral and EEG data.

Our study had three specific goals. First, we aimed to evaluate whether the effect of response monitoring on subsequent cognitive adaptation is consistent across individuals. Second, we intended to determine whether these effects are similar for both error and correct responses. Third, we sought to assess whether our results are replicable across different datasets. In addition, on an exploratory basis, we investigated whether and how different affective and motivational traits of

* Corresponding authors.
E-mail addresses: anna.belowska@doctoral.uj.edu.pl (A. Grabowska), magdalena.senderecka@uj.edu.pl (M. Senderecka).

[1] Shared senior authorship.

https://doi.org/10.1016/j.neuroimage.2025.121349
Received 13 January 2025; Received in revised form 18 April 2025; Accepted 25 June 2025
Available online 22 July 2025

individuals influence the relationship between response monitoring and adaptive behavior.

1.1. Post-error slowing as a behavioral adaptation: Theories and insights from computational modeling

Post-error slowing describes the tendency to slow down on trials following errors (Laming, 1968, 1979; Rabbitt, 1966, 1979; Rabbitt and Rodgers, 1977). Since post-error slowing often coincides with increased response accuracy (Laming, 1968; Danielmeier et al., 2011; Marco-Pallarés et al., 2008), it has been initially assumed to reflect an adaptive behavioral adjustment to prevent future errors (Allain et al., 2009; Rabbitt, 1966). However, some researchers reported that when the interval between responses from the subsequent trials is short, sometimes accuracy is unchanged, or even reduced (see Danielmeier and Ullsperger, 2011 for a review; see also Fiehler et al., 2005; Notebaert et al., 2009; Rabbitt and Rodgers, 1977; Jentzsch and Dudschig, 2009). These findings suggest that post-error slowing might be a maladaptive rather than an adaptive process that reflects impaired cognitive abilities on the next trial, e.g., perceptual or attentional distraction. In 2018, Wessel proposed an adaptive orienting theory, which suggests that an erroneous response triggers an initial automatic global inhibition of motor and cognitive processes, followed by error-specific adaptive control. If an error was caused by premature responding, adaptation requires heightening response caution (Botvinick et al., 2001; Rabbitt and Rodgers, 1977). Conversely, if an error occurred because of distraction, focusing on task-relevant features would be a more adaptive strategy (Wessel, 2018; Wessel and Aron, 2017).

Advancing this debate, Dutilh et al. (2012) distinguished five competing explanations for post-error slowing: (1) heightened response caution (e.g., Botvinick et al., 2001; Rabbitt and Rodgers, 1977), (2) a tendency to avoid repeating the recently erroneous response (Laming, 1968, 1979), (3) reduced trial-to-trial variability in bias towards one or the other response, indicating reduced processing of stimulus-unrelated information (Laming, 1968, 1979), (4) attentional distraction (Notebaert et al., 2009), and (5) delayed response initiation due to irrelevant factors such as emotional reactions (Rabbitt and Rodgers, 1977). They tested these theories using a diffusion drift modeling (DDM) approach. The DDM is a computational framework used to describe the process of decision-making that conceptualizes the response process as a realization of several unobserved cognitive properties, each described by a specific parameter (Ratcliff and McKoon, 2008; Voss et al., 2004, 2013). The DDM assumes that the decision-making process starts with an initial period of non-decision time t_0 (thought to be related to encoding), followed by evidence accumulation that moves in a random walk to one of two boundary separations. The average rate of evidence accumulation (drift rate v) describes the ability to collect information and results in faster and more accurate choices; thus, v relates to a subject's ability and focus on task-relevant features. The speed–accuracy trade-off is instantiated by variation of the distance between thresholds (boundary separation a); participants can behave more cautiously by widening the boundary separation, resulting in fewer errors, though this comes at the cost of longer reaction times (RTs). Once the accumulated evidence surpasses one of the thresholds, a motor response is initiated and executed, adding time to non-decision time t_0. Based on this computational approach, Dutilh et al. found that post-error slowing is primarily associated with increased response caution and, to a lesser extent, with a tendency to avoid repeating the recent erroneous response. Thus, their findings emphasized adaptation-related explanation of post-error slowing.

1.2. Post-error slowing and response monitoring: Neurophysiological perspective

While the post-error slowing explanations tested using the DDM approach by Dutilh et al. may align with behavioral data, they do not provide insights into the neural basis of the cognitive processes underlying adaptation. These processes are thought to be revealed through EEG data at the single-trial level.

Mattes et al. (2022) were the first to apply the DDM model to both behavioral and EEG data to investigate the underlying neural mechanisms responsible for behavioral adaptation following response execution. They found evidence that amplitudes of two well-established biomarkers of response monitoring, error-related negativity (ERN, Gehring et al., 1993, also called error negativity, Ne; Falkenstein et al., 1991) and correct-related negativity (CRN Vidal et al., 2000, 2003, also called correct negativity, Nc), are associated with several cognitive processes that lead to behavioral adaptation in decision-making tasks.

ERN and CRN are event-related potential (ERP) components that appear immediately after an erroneous or correct response, respectively, typically within 100 ms from response onset. ERN is characterized by a pronounced negative deflection in the ERP waveform over mediofrontal scalp regions, whereas CRN has a similar topography (Luu et al., 2000; Vidal et al., 2000), but exhibits a smaller amplitude. Both components presumably have the same source in the anterior cingulate cortex (ACC Roger et al., 2010), though caution must be taken when interpreting source localization results in EEG (Nunez et al., 2019). More recent studies using intracerebral recordings have indicated that both components may also originate from the supplementary motor area (Bonini et al., 2014) or the pre-supplementary motor area (Fu et al., 2019). Several competing functional theories aimed at explaining the mechanism of ERN associate variability in its amplitude with monitoring of current (erroneous) behavior and providing an early warning that behavioral adjustment is necessary. These theories interpret ERN as a regulatory signal, e.g., signal of the increase in attentional control (van Noordt et al., 2015) or negative reinforcement learning signal (Holroyd and Coles, 2002), or evaluatory signal, e.g., early evaluative stage in executive control (Weinberg et al., 2016).

Fitting the DDM to their data, Mattes et al. (2022) confirmed the idea that ERN and CRN are related to adaptive processes that aim to prevent future errors. They found that more negative amplitudes of ERN in the previous trial were associated with increased response caution, greater focus on task-relevant features, and delayed response initiation. These results align with other studies that have also reported an association between more pronounced ERN amplitudes and increased post-error slowing at the single-trial level (Park et al., 2024; Fischer et al., 2016; Wessel and Ullsperger, 2011; Debener et al., 2005; Gehring et al., 1993), reinforcing the idea that ERN reflects the cognitive processes involved in initiating control mechanisms for adaptation.

However, the connection between the cognitive processes involved in initiating control and subsequent adaptation is not considered to be direct. For example, within the model proposed by Weinberg et al. (2016), the relationship between evaluation (i.e., ERN) and compensation (i.e., post-error behavioral adjustments) may depend on intermediate processes and be mediated by various internal and external factors. This perspective aligns with other theories of response monitoring, such as the reinforcement learning theory. Hence, the connection between ERN and adaptation might not be observed consistently across all individuals but exhibit considerable variability in strength and direction. While existing studies demonstrated the association between response monitoring and subsequent adaptation, they provide limited insights into inter-individual variability of this association. In the current study, we aimed to investigate this inter-individual variability, and how external and internal factors, such as motivational and affective traits or speed–accuracy instructions, influence the relationship between neural markers of response monitoring and adaptation.

Although less studied than the ERN, the presence of the CRN in correct responses suggests that the brain's performance monitoring system is involved in evaluating all responses, not just errors (Bartholow et al., 2005). Studies have shown that more pronounced CRN amplitudes are associated with greater attention and vigilance (Matsuhashi et al., 2021; Van Noordt et al., 2016; van Noordt et al., 2015), higher motivation to respond correctly (Imhof and Rüsseler, 2019; Files et al., 2021), and higher uncertainty regarding response correctness (Pailing and Segalowitz, 2004; Scheffers and Coles, 2000). Thus, the CRN may reflect the brain's continuous monitoring of responses, whether correct or incorrect (Endrass et al., 2012).

The difference in amplitude between the ERN and CRN indicates that performance monitoring is specifically sensitive to errors, but it remains inconclusive whether the ERN and CRN reflect different neuronal processes and what the origins of their amplitude differences are (see Vidal et al., 2022, for a discussion). Mattes et al. (2022) found significant differences between ERN and CRN and their associations with behavioral adaptation, but only in some conditions. Given the mixed findings in previous research, our goal was to assess the consistency of the association between ERN and CRN and behavioral adaptation across individuals. Low consistency may indicate that the variability in reported findings could be attributed to significant inter-individual differences in the relationship between ERN, CRN, and adaptation.

1.3. Reliability of individual differences in response monitoring and behavioral adaptation

The importance of evaluating inter-individual variability in cognitive research was thoroughly discussed by Haaf and Rouder (2018) and Rouder and Haaf (2021). Haaf and Rouder differentiated between *quantitative* and *qualitative* individual differences. *Quantitative* individual differences imply that every individual in the population exhibits an effect in the same direction and variations lie solely in their magnitude. Such quantitative differences suggest a straightforward theory behind the origin of the effect: it might reflect processes that are unaffected by concurrent external or internal factors, or at least, not highly sensitive to them. In turn, *qualitative* individual differences imply that some people have negative or null effects while others have positive effects. This kind of variability suggests that the effect is dependent on the concurrent external and/or internal factors. Such factors include the use of different cognitive strategies for the same task, natural variability in learning abilities and cognitive control efficiency, as well as individual differences in affective and motivational characteristics that may influence cognitive processing (Rouder and Haaf, 2021). Importantly, this variability could also arise from scenarios where different brain processes are misidentified as the same across individuals. Proposed explanatory theories have to account for this variability and identify its sources.

If individual differences in the effects of ERN and CRN on behavioral adjustments are quantitative, it would support the idea that ERN and CRN reflect processes that actively implement cognitive control and facilitate behavioral adjustments, regardless of the influence of external or internal factors (see Fig. 1, panel A). In this scenario, ERN and CRN amplitudes could serve as reliable measures of instantiated cognitive control and adaptation signals. Conversely, if these individual differences are qualitative, meaning the effects are only observed in some individuals or vary in direction, it would suggest that the relationship between ERN, CRN, and subsequent adaptation is more complex and influenced by various factors (see Fig. 1, panel B). Alternatively, qualitative individual differences may indicate that different brain processes are being misidentified in some individuals as the ERN/CRN. In those cases, ERN and CRN amplitudes would not be reliable indicators of instantiated cognitive control. In the current study, we aim to distinguish between these two scenarios, thereby contributing to the ongoing debate regarding the functional significance of ERN and CRN.

The relevance of these considerations can be illustrated using the expected value of control (EVC) model of decision-making (Frömer et al., 2021; Frömer and Shenhav, 2022). The EVC model predicts that high incentives enhance ERN amplitude, as larger rewards are expected to motivate an increase in cognitive control, thereby amplifying response monitoring. This theory, like many decision-making models, assumes quantitative rather than qualitative dynamics between response evaluation, cognitive control, and adaptation. However, since the link between ERN/CRN amplitudes and adaptation is presumably indirect, these assumptions may not always hold true. Therefore, assessing the extent to which the link between ERN/CRN amplitudes and adaptation depends on mediating factors is crucial for developing more accurate decision-making and performance monitoring models.

1.4. The current study

Adopting the framework proposed by Rouder and Haaf (2021), our analysis aims to assess the consistency of individual-level differences in the effect of ERN/CRN on behavioral adjustments. To achieve this, we adopted the methodology put forth by Mattes et al. (2022), which links neurophysiological markers of response monitoring, i.e, ERN and CRN, to indicators of subsequent cognitive adjustment derived from the response time and accuracy on the subsequent trial using DDM (see Fig. 2). Following the results of Mattes et al. (2022), we hypothesize that more pronounced ERN/CRN peak amplitudes will be associated with increased response caution and enhanced focus on task-relevant features, reflected in larger boundary separation and higher drift rates, respectively. However, we do not have explicit hypotheses regarding individual-level effects.

Our approach utilizes Bayesian hierarchical modeling to estimate population- and individual-level effects of response type (error vs. correct). In the model, ERN/CRN amplitudes from preceding trials have a linear influence on model parameters. We follow a *directed* approach in the model-based Cognitive Neuroscience nomenclature of Palestro et al. (2018) that has been used with success by Nunez et al. (2024) to find EEG-decision-making cognition relationships. We use this approach to: (1) examine the consistency of individual differences in how ERN and CRN impact drift rate and boundary separation to determine whether these differences are qualitative or quantitative, (2) assess whether ERN and CRN effects on drift rate and boundary separation differ significantly, and (3) attempt to replicate obtained results in different datasets. Additionally, in an exploratory manner, we seek to identify affective and motivational individual differences associated with ERN and CRN effects on DDM parameters, aiming to uncover potential moderators in the ERN/CRN-adaptation relationship.

To preview our findings, at a population level, we observed a robust correspondence between more pronounced ERN amplitudes and increased boundary separation, which reflects the increased response caution, and increased drift rate, which had previously been associated with stronger attentional focus on task-relevant features (Mattes et al., 2022). Despite robust population-level effects, we observed substantial variability in the individual-level effects, with distinct patterns across different datasets, especially for effect of ERN amplitude on response caution. The relationship between response-related brain activity and subsequent adaptation was limited to error trials.

2. Method

In our study, in order to assess the replicability of obtained results, we utilized four datasets: one collected in our laboratory at Jagiellonian University in Kraków, Poland (described below) and three additional flanker datasets (letter-based and digit-based, the latter divided into two different subsets based on a between-subjects manipulation) from Mattes et al. (2022).

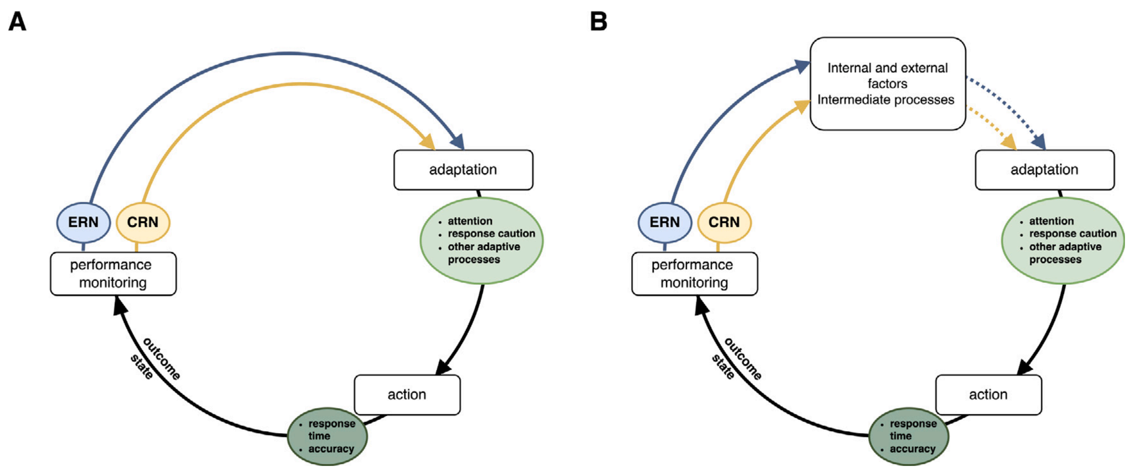

Fig. 1. Simplified cognitive control loop adapted from Ullsperger (2024) that illustrates the relationship between response-related brain activity, i.e., ERN and CRN amplitudes, and subsequent behavioral adaptation. Panel A: the link between ERN and CRN amplitudes and adaptation is shown as direct and consistent across all individuals, represented by solid lines. This indicates that the strength of this relationship is uniform and not significantly influenced by external or internal factors, implying minimal moderation. Panel B: the relationship between ERN, CRN, and adaptation is depicted as complex and moderated by various internal and/or external factors. Here, the strength of adaptation depends not only on the magnitude of the ERN and CRN signals but also on the presence and nature of these moderators, as represented by dotted lines. These moderators can vary in their influence and may even fully inhibit behavioral adaptation, highlighting the non-linear and context-dependent nature of adaptation.

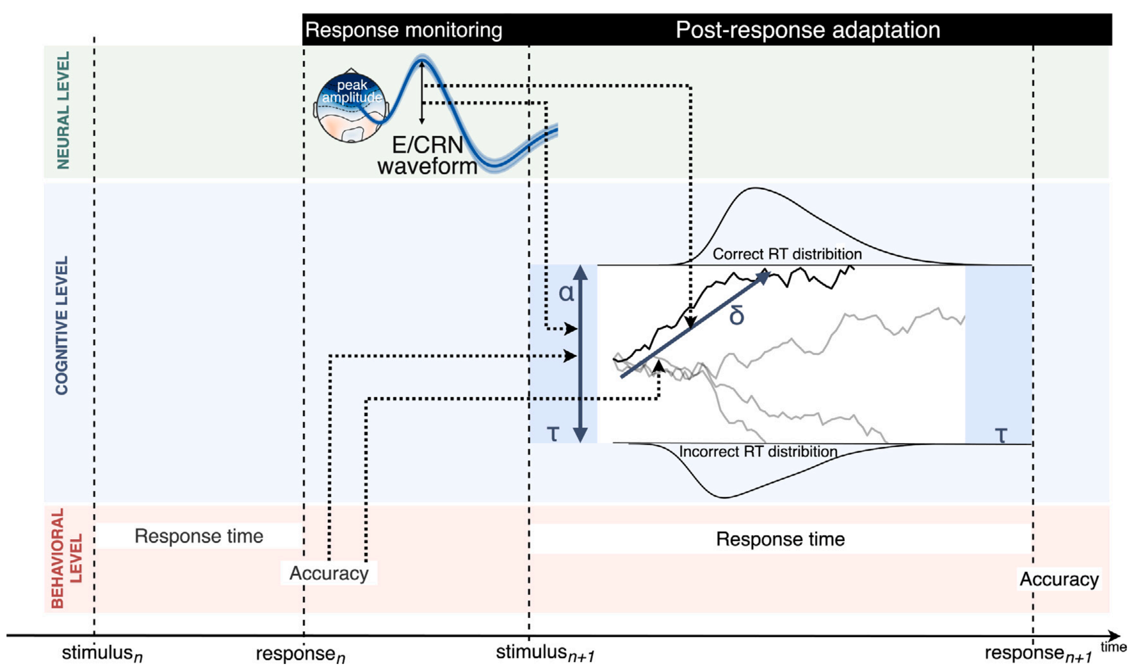

Fig. 2. A neuro-cognitive model illustrating EEG-decision-making cognition relationships across two consecutive trials. The dotted lines represent a statistical model that connects neurophysiological markers of response monitoring, i.e., the error-related negativity (ERN) and correct response negativity (CRN), to subsequent cognitive adjustments. These adjustments, quantified as drift rate δ and boundary separation α, are derived from joint response time and accuracy data using a drift diffusion model. Specifically, the peak amplitudes of the ERN/CRN and the correctness of responses in trial n are used to predict the drift rate δ and boundary separation α of the subsequent trial $n+1$.

2.1. Participants in our laboratory

The present study involved analyses of a dataset that was collected within a larger project conducted in our laboratory. We report all data exclusions, all manipulations, all measures in the study, and each step of the data analysis.

Two hundred and twenty-five volunteers (113 females, 111 males, one non-binary gender) aged 18–39 (M = 23.64, SD = 4.18) took part in the study within our laboratory. Participants were recruited from the general population via internet advertisements. All participants were healthy, free of medications, declared that they had no history of neurological or psychiatric diseases, and had normal or corrected-to-normal vision. Prior to analysis, 3 participants were excluded for the following reasons: less than six committed errors (N = 2) and low overall EEG data quality (N = 1). The final sample consisted of EEG data from 222 participants (111 females, 110 males, one non-binary gender) aged 18–39 (M = 23.73, SD = 4.14). The average number of years of education among the participants in the final sample was 15.48 (SD = 2.41). The same sample was included in the research described in Grabowska et al. (2024b,a).

2.2. Procedure and task in our laboratory

Participants received verbal and written information about the purpose and procedure of the study. The study was performed following the Declaration of Helsinki (BMJ, 1996). The protocol was approved by the Research Ethics Committee at the Philosophical Faculty of Jagiellonian University in Krakow, Poland. All participants provided written informed consent and were monetarily compensated for their time.

Participants completed (within a larger set of tasks and questionnaires) a speeded, modified version of the flanker task (Eriksen and Eriksen, 1974), validated in previous research (e.g., Fiehler et al., 2005; Ullsperger and von Cramon, 2001), see Fig. 3. In each trial, participants were presented with a fixation mark for 600–800 ms at the center of the screen, followed by four flanker arrows for 680 ms (two above and two below the center). The arrows were 0.6° tall and 1.5° wide in degrees of visual angle, measured at a distance of 80 cm from the screen (approximate distance of participants from the screen). The target arrow, presented between the four flanker arrows, appeared for 600 ms with a delay of 80 ms from the flanker arrows' onset. We used an asynchronous presentation of flankers and target because it has been shown to reliably elicit two response activations in rapid succession (one triggered by the flankers and another by the target) at the cortical level, even though typically only one of these results in an overt motor response (Kopp et al., 1996). This effect intensifies response conflict and its internal monitoring, making this design particularly suitable for studies focused on ERN/CRN, as confirmed by previous research in the field (Danielmeier et al., 2009; Endrass et al., 2008; Riesel et al., 2011). In 70% of the trials (210 trials), the flankers pointed in the same direction as the target (congruent trials), while in the remaining 30% of the trials (90 trials), they pointed in the opposite direction (incongruent trials). Congruent and incongruent trials appeared in random order. The imbalance between congruent and incongruent trials was based on findings from previous research (Bartholow et al., 2005; Gratton et al., 1992), which indicated that a higher probability of congruent compared to incongruent trials elicits a greater number of errors and increased engagement of response conflict monitoring processes.

Participants were instructed to respond as quickly and accurately as possible to the direction of the target arrow. A left-pointing target arrow required a left-hand response, while a right-pointing target arrow required a right-hand response. Responses could be made during the 600 ms presentation of the flankers and target, as well as up to an additional 600 ms after their disappearance. No explicit instructions regarding immediate corrections for errors were provided to participants. Each correct response was followed by symbolic feedback on response speed (a smiling or sad face), presented 1000 ms after the button press, indicating whether the answer was fast enough. The feedback was displayed for 600 ms, followed by an intertrial interval of 600–800 ms. The response time pressure was dynamically adjusted by an adaptive algorithm based on the participant's performance, aiming to optimize the error rate and reduce drop-outs due to a low number of error trials. The algorithm took into account the last 20 reaction times (RTs), setting the threshold at the median of these 20 RTs, considering only correct reactions. This approach ensured a robust dataset with sufficient error trials for reliable analysis, while also reducing participant dropouts due to ceiling effects in accuracy. The precise code for the algorithm is available on GitHub (https://github.com/filyp/cognitive_tasks/blob/main/classes/procedures/flanker_task/feedback.py). No feedback was provided following erroneous responses to enhance internal monitoring. All participants were asked to restrict blinking and body movements as much as possible during the EEG recording. The task was implemented using PsychoPy software (Peirce, 2007). Fig. 3 presents an outline of the flanker task design.

The task comprised one training block of 20 trials and five experimental blocks of 60 trials each. After each block, participants had the opportunity to relax briefly before the next block began.

2.3. Electrophysiological recording and data pre-processing

To ensure the comparability of our results with those obtained from the datasets of Mattes et al. (2022), we followed Mattes et al.'s preprocessing and trial selection procedures as closely as possible. The deviations form the original pre-processing procedure are summarized in Appendix A.

The experiment in our laboratory was conducted by trained researchers in a sound-attenuated room. The EEG signal was continuously recorded at 64 silver/silver-chloride (Ag/AgCl) active electrodes at a sampling rate of 256 Hz using the BioSemi Active-Two system with Common Mode Sense (CMS) and Driven Right Leg (DRL) electrodes, which drive the average potential across all electrodes as closely as possible to amplifier zero. The horizontal and vertical electro-oculograms (EOGs) were monitored using four additional electrodes placed above and below the left eye and in the external canthi of both eyes after adequate skin preparation.

The EEG pre-processing was performed using the Python programming language package MNE-Python (Gramfort et al., 2013). The signal was re-referenced offline to the average of the left and right mastoid electrodes; then a DC detrend correction was applied to the continuous EEG signal. Data were further epoched around the response onset (−100 to 900 ms) and were baseline corrected to the average pre-response activity from −100 to 0 ms. Ocular artifact correction was performed with the algorithm by Gratton et al. (1983). Data were then re-baselined to repair baseline violated by ocular correction. Epochs in which the EEG signal at the FCz or Cz site were greater than ± 150 µV were removed.

To assess the quality of EEG data, we estimated the subject-level internal consistency of ERN and CRN amplitudes. Internal consistency was calculated using a subject-level dependability coefficient (see Clayson and Miller, 2017; Clayson et al., 2021), as described in Eq. (1):

$$\phi_{jk} = \frac{\sigma^2_{(p)}}{\sigma^2_{(p)} + \sigma^2_{(i)jk}/n_{(i)jk}} \tag{1}$$

where $\sigma^2_{(p)}$ is between-subject variance, $\sigma^2_{(i)jk}$ is person-specific within-subject variance, and $n_{(i)jk}$ is the person-specific number of trials i for a given person j from group k.

We followed a trial selection procedure, as employed by Mattes et al.. Firstly, we excluded all trials from the analyses with a response time (RT) greater than 1000 ms. We then analyzed trials that followed the pattern "$CCXP$": where C indicates a correct response, X is a reference trial providing response type (erroneous/correct) and ERN/CRN amplitude predictors, and P indicates a trial aimed at observing cognitive adaptation. Limiting X trials to those preceded by two correct answers ensured that results were not biased by successive errors (Schiffler et al., 2017). Furthermore, we excluded P trials with a log-transformed RT outside the population mean ±three standard deviations (Voss et al., 2015). On average, we excluded 23.76% of trials per participant (SD = 9.88) from the analysis. After trial selection, the average number of correct trials per participant included in the analysis was 202.17 (*SD* = 37.46), while the average number of erroneous trials per participant included in the analysis was 24.95 (*SD* = 8.26).

2.4. Mattes et al. datasets

To assess the replicability of our findings across datasets, we used the data from the flanker task with baseline, speed, and accuracy conditions by Mattes et al. available at the Open Science Framework (OSF) platform (https://osf.io/8nbd4/). Detailed descriptions of the flanker task used, data pre-processing methods, and the single-trial ERP estimation procedure can be found in the paper by Mattes et al. (2022).

2.5. Statistical analyses and procedure

In the main analysis we included multiple predictors to estimate different neurocognitive effects. The overview of the predictors, along with the effect parameters, is show in Table 1. To estimate the effects of ERN and CRN separately, we summed the effects of ERN/CRN

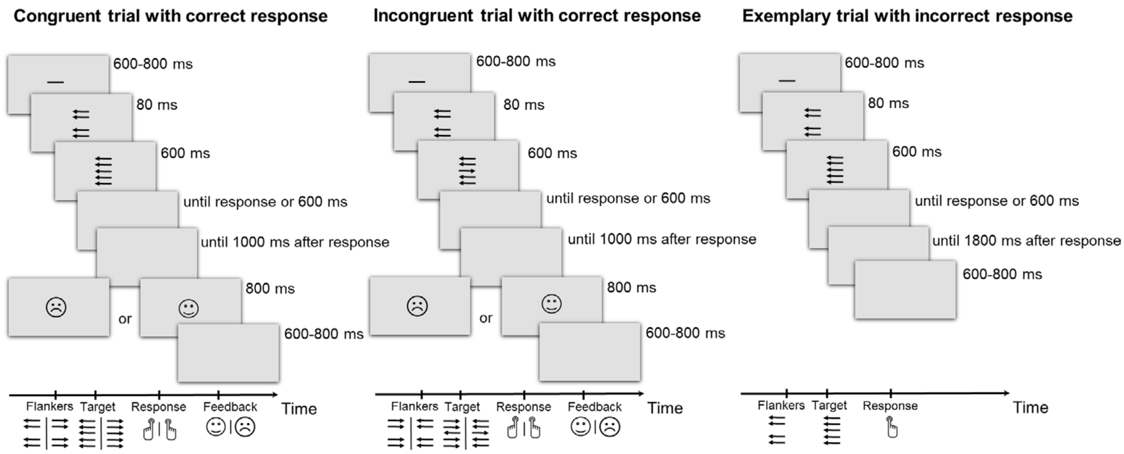

Fig. 3. Flanker task. Participants were required to respond according to the direction of the central arrow by pressing the left or right button. The left panel presents a congruent trial with a correct response, the middle panel shows an incongruent trial with a correct response, and the right panel illustrates an exemplary trial with an incorrect response.

Table 1
Summary of the predictors and effect parameters from the main analysis.

Aim	Predictor	Fixed	Random
Estimate association between ERN/CRN amplitude and the subsequent adaptation	Single-trial ERN/CRN peak amplitudes from the preceding trial	$\mu_{(e)}$	$\theta_{(e)j}$
Estimate association between response type and the subsequent adaptation	Response type (error vs. correct) indicator from the preceding trial	$\theta_{(r)}$	–
Estimate the difference between ERN and CRN associations with cognitive adaptation	Single-trial ERN/CRN peak amplitudes multiplied by response type indicator	$\mu_{(er)}$	$\theta_{(er)j}$
Control for the effect of condition	Condition type indicator of the current trial (congruent vs. incongruent)	$\mu_{(c)}$	$\theta_{(c)j}$
Control for the interaction effect between ERN/CRN amplitude and condition	Single-trial ERN/CRN peak amplitudes multiplied by condition type indicator	$\theta_{(ec)}$	–
Control for the interaction effect between condition and response type	Condition type indicator multiplied by response type indicator	$\theta_{(rc)}$	–
Control for the interaction effect between condition, response type, and ERN/CRN amplitude	Single-trial ERN/CRN peak amplitudes multiplied by response type indicator and condition type indicator	$\theta_{(erc)}$	–
Estimate association between ERN amplitude and the subsequent adaptation	Summed effects of ERN/CRN peak amplitude $\theta_{(e)j}$ with the ERN/CRN × response type interaction effects $\theta_{(er)j}$ for the response type = incorrect	$\mu_{(EN)}$	$\theta_{(EN)j}$
Estimate association between CRN amplitude and the subsequent adaptation	Summed effects of ERN/CRN peak amplitude $\theta_{(e)j}$ with the ERN/CRN × response type interaction effects $\theta_{(er)j}$ for the response type = correct	$\mu_{(CN)}$	$\theta_{(CN)j}$

Note. i indicates trial number, and j indicates the individual to whom the trial belongs. Thus, all effects with j index indicate random slopes. c, e, and r stand for condition, ERN/CRN amplitude, and response type, respectively.

amplitude $\theta_{(e)j}$ with the ERN/CRN-response type interaction effects $\theta_{(er)j}$ for each model result.

The amplitude of response-related brain activity from the preceding trial, i.e., ERN/CRN, was standardized in a participant-wise manner, by subtracting the mean and then dividing by the standard deviation for each participant separately. The response type from preceding trial, i.e., correct/incorrect response type, and the condition type, were contrast coded (error = 1 vs. correct = -1; congruent = 1 vs. incongruent = -1, respectively).

All estimated models were developed using Bayesian hierarchical modeling (Lee and Newell, 2011; Rouder and Lu, 2005; Vandekerckhove et al., 2011) following a trend in model-based cognitive neuroscience (e.g. Frank et al., 2005; Nunez et al., 2017), enabling estimation of effects at both the population and individual levels. In the main analysis we nested models at the participant level and included random intercepts and random slopes for condition type, ERN/CRN amplitude, and the interaction between ERN/CRN and response type (see Table 1). This approach allows intercepts and the effects of condition type, ERN/CRN amplitude, and the interaction between ERN/CRN and response type to vary across participants, thereby allowing us to examine individual differences. The remaining predictors were treated as fixed effects and were estimated at the population level.

All Bayesian models were created in Stan (Stan Development Team, 2024) through CmdStanPy (Stan Development Team, 2021). Stan is a probabilistic programming language for statistical inference. It employs an advanced dynamic Hamiltonian Monte Carlo algorithm (Betancourt, 2018, 2016) based on a variant of the No-U-Turn sampler (NUTS; Hoffman and Gelman, 2011), which generally offers greater efficiency compared to traditional MCMC samplers used in other probabilistic programming languages.

All models were fitted with the following parameters: number of chains = 6, adapt delta = 0.99, maximal tree depth = 12. To assess the convergence and efficiency diagnostics, we calculated split-$\hat{R}$ (Gelman–Rubin; Gelman et al., 2015) and effective sample sizes for each of the estimated parameters. Basic statistical analyses were conducted using Scikit-learn (Pedregosa et al., 2011), NumPy (Harris et al., 2020), and ArviZ (Kumar et al., 2019) Python programming language packages.

The code to reproduce all analyses is available at https://github.com/abelowska/jointErrorCMD under the MIT license. The dataset, consisting of EEG, behavioral and questionnaire data, along with the Supplementary Materials, is available on the OSF platform (https://osf.io/jszyh/); we also provide all model outputs to ensure the analyses are fully reproducible.

2.5.1. Non-cognitive models

To ensure the robustness of our results, we employed more standard statistical models in addition to the DDM. To estimate the effect of ERN/CRN on subsequent behavioral adaptation, we regressed the RTs on the ERN/CRN amplitudes and response times using multilevel linear regression, while controlling for the effect of condition, as described in Table 1. Similarly, we analyzed the effect of ERN/CRN and response type on accuracy, but employed a multilevel logistic regression instead of a linear regression. The full models' specification is in Appendix B.

2.5.2. Diffusion model analysis

Following Mattes et al., we modeled error vs. correct responses. Thus, the response boundaries in our model represented error and correct responses (instead of the two response options). We fixed the bias parameter at 0.5α, assuming that the evidence accumulation process towards correct or error responses had no theoretical bias towards either response. We used a simplified four-parameter Wiener distribution to model RT and accuracy. Thus, the inter-trial variability parameters for the starting point, the drift rate, and the non-decision time were fixed to 0 (following recommendations by Lerche and Voss, 2016; Boehm et al., 2018). To estimate the effect of ERN/CRN on subsequent cognitive adaptation, we regressed the drift rate and boundary separation parameters on the predictors described in Table 1. For example, the decomposition of the drift rate was as follows:

$$\delta_{(ij)} = \theta_{(\delta)j} + \theta_{(c)j} * \mathrm{c}_i + \theta_{(e)j} * \mathrm{e}_i + \theta_{(r)} * \mathrm{r}_i + \theta_{(er)j} * \mathrm{e}_i * \mathrm{r}_i + \theta_{(ec)} * \mathrm{e}_i * \mathrm{c}_i + \theta_{(rc)} * \mathrm{r}_i * \mathrm{c}_i + \theta_{(erc)} * \mathrm{e}_i * \mathrm{r}_i * \mathrm{c}_i$$

where i indicates trial number and j indicates the individual to whom the trial belongs; c_i is the condition type at trial i; r_i is the response type preceding trial i; e_i is a ERN/CRN response-related amplitude preceding trial i. θ denotes effects listed in Table 1 on the drift rate (c, e, r stand for condition, ERN/CRN amplitude, and response type, respectively). The decomposition of the boundary separation parameter was identical to the decomposition of the drift rate. A detailed and complete model specification of the DDMs can be found in Appendix C.

Our primary aim was to assess the reliability of individual differences in the effects of ERN and CRN on drift rate and boundary separation. Thus, for each participant, we calculated Bayes Factors (BFs) using the Savage–Dickey ratio (BF; Nunez et al., 2024; Wagenmakers et al., 2010; Dickey, 1971). For the fixed effects, we interpreted BF values as follows: BF above 5 indicates an evidence for an effect, BF between 5 and .20 suggests insufficient evidence to determine the presence of the effect, and BF lower than .20 indicates strong evidence for the null effect. Since random effects typically have weaker evidence due to a smaller amount of data, we interpreted BF values for random effects as follows: a BF above 3 indicates evidence for an effect, a BF between 3 and 0.33 suggests insufficient evidence to determine the presence of an effect, and a BF lower than 0.33 indicates strong evidence for the null effect.

Models comparison: To determine if the main model with effects on both drift rate and boundary separation provides superior explanations for variability in response times and accuracies, we constructed additional DDM models, namely: (1) with only effects on the boundary separation, (2) with only effects on the drift rate, and (3) without response type and CRN/ERN effects (though each model type always included condition type effects on boundary and drift). We then evaluated these models using the Watanabe–Akaike information criterion (WAIC; Watanabe, 2010).

In the drift diffusion model, the diffusion coefficient parameter quantifies noise in the evidence accumulation process, influencing both response time and decision accuracy. Since boundary separation, drift rate, and the diffusion coefficient are not jointly identifiable (see Nunez et al., 2023), the diffusion coefficient is typically assumed constant, with other parameters estimated relative to it. However, if the diffusion coefficient varies across individuals, this variability may be misattributed to drift rate or boundary separation in models where it is fixed. To address this, we also fitted a model allowing the diffusion coefficient to vary. Details and results of this alternative model are provided in Appendix D.

Replicability: We aimed to assess the replicability of our results across different datasets. To achieve this, we used no-instruction (Base), speed-instruction, and accuracy-instruction data from Mattes et al. (2022), fitting these datasets to our model.

2.5.3. Exploratory correlation analysis

In an exploratory analysis, we aimed to identify internal and external factors associated with ERN and CRN effects on DDM parameters, to uncover potential moderators in the ERN/CRN-adaptation relationship.

Moderator selection was based on a literature review. We included (1) the number of erroneous and correct trials used in the DDM analysis, as prior research suggests associations between trial counts and estimates of boundary separation and drift rate (see Lüken et al., 2023); (2) the number of errors committed during the task, since evidence indicates that habituation to errors may reduce ERN amplitude over time (Boksem et al., 2006; Fischer et al., 2017; Senderecka et al., 2018); (3) ERN and CRN consistency calculated as dependability coefficient; and (4) self-report trait-related individual differences. From the set of questionnaires (Polish adaptations or Polish translations made using a forward–backward translation protocol) completed by participants, the following were selected: State-Trait Anxiety Inventory (Spielberger et al., 1970, 2006), trait subscale, to measure anxiety; Behavioral Inhibition System Scale (Carver and White, 1994; Müller and Wytykowska, 2005) to measure behavioral inhibition; Intolerance of Uncertainty Scale — 12 (Carleton et al., 2007), inhibitory and prospective subscales, to measure intolerance of uncertainty; Obsessive-Compulsive Inventory-Revised (Foa et al., 2002; Huppert et al., 2007) to measure obsessive-compulsive symptoms severity; Penn State Worry Questionnaire (Meyer et al., 1990; Janowski, 2007) to measure worry; and Need For Closure Scale (Webster and Kruglanski, 1994; Kossowska, 2003), decisiveness subscale, to measure decisiveness. All these traits are frequently linked with more (see Weinberg et al., 2015; Riesel et al., 2023, for a review) or less (Senderecka et al., 2018) pronounced ERN amplitudes. Total scores for all self-report measures were calculated according to the authors' instructions and normalized by the number of questions.

The correlation analysis examined relationships between the variables described above and individual-level ERN and CRN effects on boundary separation and drift rate. If a variable was significantly correlated with the effect of ERN/CRN on boundary separation or drift rate, it was interpreted as a potential moderator of this effect. The correlation analysis was conducted in a Bayesian framework using the BayesFactor package in R, which allowed for the quantification of evidence both for and against the presence of an association.

3. Results

Unless stated otherwise, all the analyses and results presented in the sections below refer to the current dataset (N = 222).

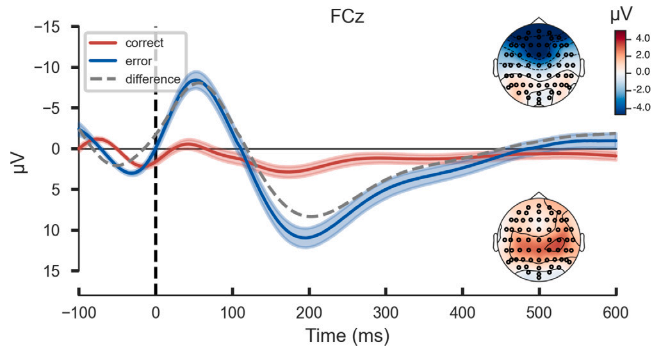

Fig. 4. Grand average waveforms from all single trials included in the analysis at electrode site FCz for the error-related negativity (ERN, blue), correct-related negativity (CRN, red), and difference waveform (dashed, gray). The headmaps show the topographies of response-related brain activity depicting the mean activity in the time window from 0 to 100 ms after erroneous (upper headmap) and correct (lower headmap) responses.

3.1. Behavioral measures

After trial selection, the average RT per participant for erroneous responses was 236 ms (SD = 27 ms), whereas for correct responses it was 288 ms (SD = 32 ms). As expected, participants responded significantly faster on erroneous trials relative to correct trials ($t(221) = -39.98$, p <.001). The average RT per participant for correct responses in congruent trials was 265 ms (SD = 30 ms), whereas in incongruent trials it was 372 ms (SD = 33 ms). We observed a congruency effect, as participants responded significantly faster on congruent trials compared to incongruent trials ($t(221) = 69.15$, p <.001). The average performance in the task, calculated as the proportion of correct responses, was 0.88 (SD = .05).

3.2. ERN and CRN amplitudes

There was a significant effect of response type (erroneous vs. correct) on EEG peak amplitude at FCz in the selected time window of 0–100 ms after the response onset ($t(221) = -20.81$, p <.001). The average ERN peak amplitude was −15.44 μV (SD = 7.80 μV). The average CRN peak amplitude was −6.41 μV (SD = 4.11 μV). Note that these means and standard deviations were computed across individual subjects' averages, while the main analysis described below was performed on the single-trial data. The grand average waveforms from all single trials included in the analysis for ERN and CRN are shown in Fig. 4.

The within-subject variance of ERN amplitudes across the 222 participants ranged from 4.40 to 18.44 μV^2 (M = 10.41 μV^2; SD = 2.50 μV^2). Internal consistency of ERN amplitudes calculated as a *dependability coefficient* (see Method section), across the 222 participants ranged from .70 to .98 (M = .92; SD = .048). The within-subject variance of CRN amplitudes ranged from 5.19 to 13.94 μV^2 (M = 8.98 μV^2; SD = 1.67 μV^2). Dependability coefficient of CRN amplitudes across the 222 participants ranged from .93 to .99 (M = .98; SD = .011).

3.3. Non-cognitive models

We explored the impact of ERN/CRN on subsequent behavioral adaptation by regressing RTs and accuracies on the predictors outlined in Table 1. The models were nested at the participant level and included random intercepts and random slopes for condition type, ERN/CRN amplitude, and the interaction between ERN/CRN and response type. All other effects were treated as fixed. Both response time and response accuracy models exhibited satisfactory convergence and efficiency statistics: all parameter achieved $\hat{R}$ less than 1.05 and effective sample sized above 100. The detailed results on the model convergence are available on OSF. The results of the regression models are summarized in Table 2.

The hierarchical Bayesian model revealed evidence for effects on response times from both the EEG amplitude $\mu_{(e)}$ (b = −2.96, 95% HDI [−3.82, −2.14], BF = 1.29×10^{80}) and response type $\theta_{(r)}$ of the previous trial. The Bayes Factors are large and represent considerable evidence from the data for these effects. Consequently, larger EEG amplitudes and error commissions in the preceding trial were associated with slower response times. There was no difference between effects of ERN and CRN on response times, with the Bayes Factor providing evidence in favor of the null effect (BF = 2.64×10^{-3}). Additionally, incongruent trials were characterized by longer response times, as expected.

For response accuracy, the analysis yielded evidence for fixed effects of the EEG amplitude $\mu_{(e)}$ and response type $\theta_{(r)}$ of the previous trial. This effect was also qualified by an EEG amplitude and response type interaction effect $\mu_{(er)}$. Thus, only more pronounced ERN amplitudes in the preceding trials were significantly associated with a higher probability of a correct responses (b = −.25, 95% HDI [−.39, −.10], BF = 326.01). There was no relationship between the CRN amplitude in the preceding trial and response accuracy (b = −.02, 95% HDI [−.09, .06], BF = .21). Congruent trials showed a higher probability of yielding a correct response. Notably, on the population level, the Bayes Factor for the ERN amplitude effect is higher for the response-time model than for the accuracy model, indicating stronger evidence for the effect of ERN amplitude from the previous trial on response time compared to response accuracy.

We investigated the variability among participants in the effects of EEG amplitude from the preceding trial on both response time and response accuracy. Fig. 5 visualizes the variability between participants in the effects of ERN and CRN amplitudes on response time and response accuracy. Green individual credible intervals correspond to evidence for an effect, orange intervals represent evidence against an effect, and gray intervals represent inconclusive evidence. For response time, individual effects of EEG amplitude and interaction effects between EEG amplitude and response type were not estimated with enough precision. While participants' posterior means were close to the population posterior mean, the wide posterior distributions of the random effects, along with the estimated Bayes Factors, indicated that this effect is not robust and cannot be observed at the individual level, in contrast to the population level.

For response accuracy, only some participants exhibited reliable effects of EEG amplitude and its interaction with response type. Further, the effect of EEG amplitude appears to be primarily driven by the ERN, which aligns with the population-level findings. For the majority of individuals, the Bayes Factors were uninformative, indicating substantial uncertainty in determining the presence of the effect at the individual level. While the effect of ERN amplitude of the preceding trial on current trial response accuracy likely exists, we cannot confidently identify which individuals show the effect.

Thus, although the population-level effect of ERN amplitude on response time is stronger than on response accuracy, the association between ERN amplitude and response time shows more variability/noise

Table 2
Posterior means (*b*), standard deviations of the posterior distributions (*SD*), 95% Highest Posterior Density Intervals (*HDIs*), and Bayes Factors (*BF*) of the fixed effects for response time and response accuracy.

Predictors	Response time				Response accuracy			
	b	*SD*	*HDIs*	*BF*	*b*	*SD*	*HDIs*	*BF*
(Intercept)	301.82	2.42	[297.06, 306.54]	n.a	3.00	0.07	[2.87, 3.13]	n.a
Condition type $\mu_{(c)}$	−29.05	0.85	[−30.77, −27.41]	$>10^{308}$	2.02	0.06	[1.90, 2.14]	$>10^{308}$
EEG amplitude $\mu_{(e)}$	−2.96	0.43	[−3.82, −2.14]	1.29×10^{80}	−0.13	0.04	[−0.21, −0.05]	31.28
Response type $\theta_{(r)}$	−10.90	0.54	[−11.95, −9.85]	$>10^{308}$	−0.14	0.05	[−0.24, −0.05]	25.27
EEG amplitude × Response type $\mu_{(er)}$	0.27	0.44	[−0.60, 1.12]	2.64×10^{-3}	0.12	0.04	[0.04, 0.19]	13.98
EEG amplitude × Condition type $\theta_{(ec)}$	1.52	0.40	[0.72, 2.30]	1.01	0.00	0.04	[−0.09, 0.07]	0.21
Response type × Condition type $\theta_{(rc)}$	1.62	0.53	[0.59, 2.67]	0.38	0.25	0.05	[0.15, 0.34]	7.36×10^{17}
EEG amplitude × Response type × Condition type $\theta_{(erc)}$	−0.72	0.40	[−1.51, 0.06]	0.01	0.08	0.04	[−0.01, 0.16]	1.05
ERN amplitude $\mu_{(EN)}$	−3.23	0.77	[−4.66, −1.68]	449.59	−0.25	0.07	[−0.39, −0.10]	326.01
CRN amplitude $\mu_{(CN)}$	−2.69	0.42	[−3.51, −1.86]	1.35×10^{53}	−0.02	0.04	[−0.09, 0.05]	0.21
Intercept variability σ	34.90	1.7	[31.70, 38.35]	n.a	0.68	0.05	[0.59, 0.77]	n.a
Condition type variability $\sigma_{(c)}$	9.83	0.58	[8.78, 11.04]	n.a	0.54	0.04	[0.45, 0.63]	n.a
EEG amplitude variability $\sigma_{(e)}$	2.39	0.56	[1.31, 3.47]	n.a	0.09	0.04	[0.01, 0.16]	n.a
EEG amplitude × Response type variability $\sigma_{(er)}$	2.80	0.48	[1.82, 3.69]	n.a	0.05	0.03	[0.01, 0.12]	n.a

Note. *c*, *e*, and *r* denote condition, ERN/CRN amplitude, and response type, respectively. Detailed descriptions of the predictors are provided in Table 1. Bayes Factor was calculated as Savage–Dickey ratio. *BF* above 5 indicates an evidence for an effect, *BF* between 5 and . 20 suggests insufficient evidence to determine the presence of the effect, and *BF* lower than . 20 indicates strong evidence for the null effect. *BF* $>10^{308}$ indicates a BF value exceeding the double-precision floating-point limit. Note that EEG, ERN, and CRN amplitudes, as well as response types, come from the preceding trial.

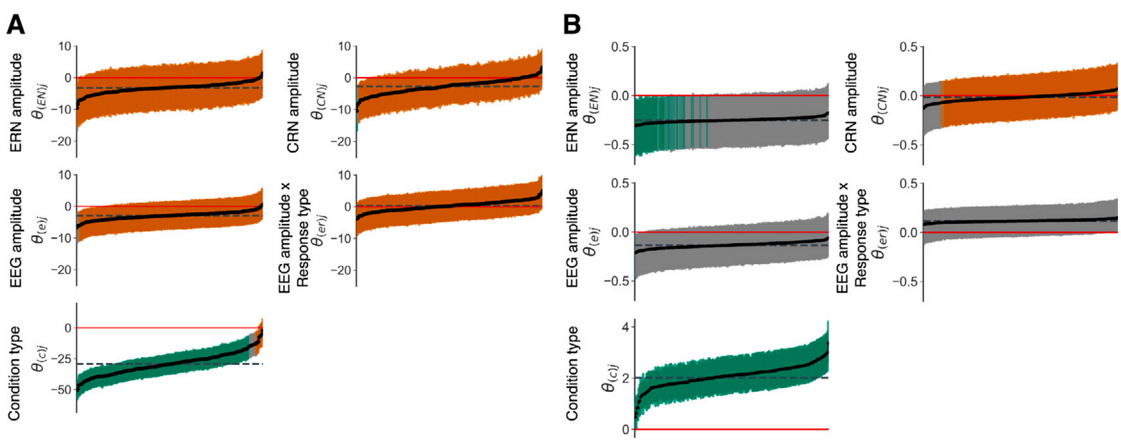

Fig. 5. Individual EEG amplitudes effects ($\theta_{(e)j}$) and their interactions with response type ($\theta_{(er)j}$), along with the effects of ERN ($\theta_{(EN)j}$), CRN ($\theta_{(CN)j}$), and condition type ($\theta_{(c)j}$) on the response time (panel A) and response accuracy (panel B). The effects are sorted by smallest to largest posterior means for each subplot. The dashed horizontal line denotes the population-level posterior means of the effects. Credible intervals (CrIs) are colored as follows: orange for Bayes Factors less than 0.33, gray for Bayes Factors between 0.33 and 3, and green for Bayes Factors greater than 3. Note that EEG, ERN, and CRN amplitudes, as well as response type, come from the preceding trial.

across participants. In contrast, while the effect on response accuracy is smaller, it is more stable and therefore more consistently detected across individuals.

3.4. Diffusion model analyses

3.4.1. Main analysis

In the main analysis, we estimated the effects of EEG amplitudes and response types on diffusion model parameters representing adaptation, specifically boundary separation and drift rate. We controlled for the effect of condition, as detailed in Table 1. Notably, the key effects for the analysis, EEG amplitudes and the interaction between EEG amplitudes and response type, were treated as random effects. This approach enabled us to assess the consistency of the effects of ERN and CRN amplitudes on adaptation across participants. A summary of the estimated fixed effects can be found in Table 3.

The main model demonstrated satisfactory convergence and efficiency statistics. Most parameters achieved $\hat{R}$ values below 1.01 and effective sample sizes exceeding 800. The highest $\hat{R}$ was 1.07, and the lowest effective sample size was 81.86, observed for the between-subject variability parameter of the EEG amplitude and response type interaction effect $\theta_{\delta er(j)}$ effect. Detailed results on model convergence are available on OSF.

At the population level, the effect of EEG amplitude and the interaction effect of EEG amplitude and response type were associated with both the boundary separation and the drift rate. Thus, more pronounced ERN amplitudes in the preceding trial were associated with higher boundary separations (b = −0.05, 95% HDI [−0.07, −0.03], BF = 4.10×10^7) and drift rates (b = −0.16, 95% HDI [−0.24, −0.07], BF = 113.00). We observed null effects of CRN amplitude on the boundary separation (b = −0.01, 96% HDI [−0.02, 0.01], BF = 0.08) and the drift rate (b = 0.01, 95% HDI [−0.03, 0.05], BF = 0.05).

Figs. 6 and 7, panels A, visualize the variability between participants' effects of ERN and CRN amplitudes from the preceding trial on drift rate and boundary separation, respectively. Specifically, it shows the posterior means of the individual slopes for EEG amplitudes ($\theta_{\delta(e)j}$ and $\theta_{\alpha(e)j}$), interaction effect of EEG and response type ($\theta_{\delta(er)j}$ and $\theta_{\alpha(er)j}$), ERN amplitudes ($\theta_{\delta(EN)j}$ and $\theta_{\alpha(EN)j}$) and CRN amplitudes ($\theta_{\delta(CN)j}$ and $\theta_{\alpha(CN)j}$), along with their corresponding credible intervals (CrIs).

At the individual level, we observed considerable variability in the effects of EEG amplitudes $\theta_{(e)j}$ and the interaction between EEG

Table 3
Posterior means (*b*), standard deviations of the posterior distributions (*SD*), 95% Highest Posterior Density Intervals (*HDIs*), and Bayes Factors (*BF*) of the fixed effects for the boundary separation and the drift rate.

Predictors	Boundary separation α				Drift rate δ			
	b	*SD*	*HDIs*	*BF*	*b*	*SD*	*HDIs*	*BF*
(Intercept)	1.48	0.02	[1.44, 1.52]	n.a	4.03	0.08	[3.87, 4.19]	n.a
Condition type $\mu_{(c)}$	0.45	0.01	[0.42, 0.48]	$>10^{308}$	3.07	0.08	[2.92, 3.22]	$>10^{308}$
EEG amplitude $\mu_{(e)}$	−0.03	0.01	[−0.04, −0.02]	8.40×10^{21}	−0.07	0.02	[−0.12, −0.03]	7.54
Response type $\theta_{(r)}$	−0.09	0.01	[−0.10, −0.08]	$>10^{308}$	−0.18	0.03	[−0.24, −0.12]	1.37×10^{27}
EEG amplitude × Response type $\mu_{(er)}$	0.02	0.01	[0.01, 0.04]	24	0.08	0.02	[0.03, 0.13]	25.24
EEG amplitude × Condition type $\theta_{(ec)}$	0.00	0.01	[−0.01, 0.01]	0.03	0.06	0.02	[0.02, 0.11]	1.32
Response type × Condition type $\theta_{(rc)}$	−0.02	0.01	[−0.04, −0.01]	9.95	0.21	0.03	[0.15, 0.27]	8.23×10^{122}
EEG amplitude × Response type × Condition type $\theta_{(erc)}$	0.01	0.01	[−0.01, 0.02]	0.06	0.04	0.02	[−0.01, 0.08]	0.16
Intercept variability σ	0.26	0.01	[0.23, 0.29]	n.a	1.13	0.06	[1.02, 1.25]	n.a
Condition type variability $\sigma_{(c)}$	0.18	0.01	[0.16, 0.20]	n.a	1.02	0.05	[0.92, 1.12]	n.a
EEG amplitude variability $\sigma_{(e)}$	0.03	0.01	[0.02, 0.04]	n.a	0.04	0.02	[0.01, 0.08]	n.a
EEG amplitude × Response type variability $\sigma_{(er)}$	0.03	0.01	[0.02, 0.04]	n.a	0.04	0.03	[0.01, 0.09]	n.a
ERN amplitude $\mu_{(EN)}$	−0.05	0.01	[−0.07, −0.03]	4.10×10^{7}	−0.16	0.04	[−0.24, −0.07]	113.00
CRN amplitude $\mu_{(CN)}$	−0.01	0.00	[−0.02, 0.01]	0.08	0.01	0.02	[−0.03, 0.05]	0.05

Note. *c*, *e*, and *r* denote condition, ERN/CRN amplitude, and response type, respectively. Detailed descriptions of the predictors are provided in Table 1. Bayes Factor was calculated as Savage–Dickey ratio. *BF* above 5 indicates an evidence for an effect, *BF* between 5 and . 20 suggests insufficient evidence to determine the presence of the effect, and *BF* lower than . 20 indicates strong evidence for the null effect. *BF* $>10^{308}$ indicates a BF value exceeding the double-precision floating-point limit. Note that EEG, ERN, and CRN amplitudes, as well as response types, come from the preceding trial.

amplitude and response type $\theta_{(er)j}$ on both the boundary separation and drift rate. In contrast to the population-level effects, Bayes Factors and posterior distributions revealed that only a few individuals exhibited a significant effect of EEG amplitude on the boundary separation, while most individuals showed evidence of a null effect. Similarly, the interaction between EEG amplitude and response type was largely null across participants, suggesting minimal differences in the effects of ERN and CRN amplitudes on the boundary separation for most individuals. For the drift rate, half of the participants showed null effects of EEG amplitude or its interaction with response type, while the other half likely exhibited some effect of EEG amplitude and its interaction with response type. However, due to high uncertainty in the individual posterior distributions, the Bayes Factors remained largely uninformative.

Both ERN and CRN amplitudes showed considerable variability in individual-level effects on boundary separations. Bayes Factors revealed that while some participants had a reliable ERN amplitude effect, about half showed a null effect. For the CRN amplitude, although most participants had null effects, for a few Bayes Factors provided the evidence for the effect. These findings suggest a lack of consistency in the effects of ERN and CRN amplitudes on the boundary separations. Regarding the drift rate, all participants had a negative posterior mean for the effect of ERN amplitude, but there was considerable individual-level uncertainty, with Bayes Factors failing to provide informative evidence on which participants demonstrated the effect. Conversely, the effect of CRN amplitude remained consistently close to zero across all participants. These findings suggest a consistent, reliable absence of association between CRN amplitude and drift rate in subsequent trials, while indicating a moderately reliable, but consistent, effect of ERN amplitude across individuals.

To determine whether the main model, which incorporates the effects of response type and EEG amplitudes on both drift rate and boundary separation, provides a better explanation for variability in response times and accuracies, we compared it with several alternative models. These models included a null model with only the condition type effect, as well as models assessing the effects of response type and EEG amplitudes on either boundary separation or drift rate individually. The WAIC for the main model was -123,943.04. In comparison, the boundary separation model had a WAIC of −123,638.24, the drift rate model had a WAIC of −123,459.12, and the null model had a WAIC of −122,825.76. These results suggest that the main model provides the best explanation for the variability in response times and accuracies. A detailed results of the alternative models are available on OSF.

3.4.2. Comparison to Mattes et al.

To evaluate the replicability of our findings across different datasets, we used no-instruction (Base), speed-instruction, and accuracy -instruction data from Mattes et al. (2022). Table 4 shows the summary statistics for the datasets used.

The current dataset exhibited the highest mean dependability coefficient for both the ERN (.92) and CRN (.98). The lower mean dependability coefficient was observed in the dataset where speed was emphasized (speed-instruction; .77 and .84 for the ERN and CRN, respectively). Compared to the datasets from Mattes et al. (2022), in the current dataset the mean response times were much shorter, and the ERN and CRN amplitudes were more pronounced.

The datasets from Mattes et al. (2022) was fitted using the same DDM model as in our main analysis (see Table 1). A summary of the estimated fixed effects can be found in Table 5.

The association between EEG amplitude and the boundary separation ($\mu_{\alpha e}$) and drift rate ($\mu_{\delta e}$) was replicated in both the accuracy-instruction dataset and the no-instruction dataset. In the speed -instruction dataset, only the effect of EEG signals on the boundary separation ($\mu_{\alpha e}$) was replicated, albeit with reduced robustness. The effect of EEG signals on the drift rate ($\mu_{\delta e}$) was absent (BF = 0.08). The interaction between EEG amplitude and response type varied across datasets. It was significant in the current dataset for both boundary separation and drift rate, but these effects were replicated only in the accuracy-instruction dataset. In the speed-instruction dataset, the interaction effect of EEG amplitude and response type was replicated only for the boundary separation ($\mu_{\alpha er}$), while for the drift rate ($\mu_{\delta er}$) it was present but not robust, with inconclusive evidence from the Bayes Factor (BF = 0.41, 95% HDI [0.01, 0.11]). In the no-instruction dataset, we found evidence for the null interaction effect for both the boundary separation and drift rate.

In the current dataset, we found that more pronounced ERN amplitudes were associated with both the boundary separation and drift rate. We successfully replicated both findings only in the accuracy-instruction dataset. In the no-instruction and speed-instruction datasets, more pronounced ERN amplitudes were robustly associated only with the boundary separation. For the drift rate, while the effect of ERN appears to be present, the Bayes Factors provide inconclusive evidence for its significance (BF = 0.19 for the no-instruction dataset; BF = 0.43 for the speed-instruction dataset).

We replicated the absence of an association between CRN amplitudes and both the boundary separation and drift rate in the speed-instruction dataset and partially in accuracy-instruction dataset. In accuracy dataset, while for the drift rate Bayes Factor provided evidence

Table 4
Mean (*M*) and standard deviation (*SD*) for the behavioral and ERP measures across the datasets used in the analyses split by response type (correct vs. error).

Variable	Correct		Error		Total	
	M	*SD*	*M*	*SD*	*M*	*SD*
Current dataset (N = 222)						
EEG amplitude	−6.41	4.11	−15.44	7.80	−7.42	4.15
EEG reliability	0.98	0.01	0.92	0.05	0.97	0.01
Response time	288.01	32.21	236.83	27.99	282.48	32.52
Performance	n.a.	n.a.	n.a.	n.a.	88.46	5.36
Number of trials	202.17	37.46	24.95	8.26	227.11	30.67
No-instruction (Base) from Mattes et al. (2022) **(N = 89)**						
EEG amplitude	−0.56	0.92	−9.71	5.50	−1.28	1.07
EEG reliability	0.94	0.06	0.73	0.20	0.88	0.12
Response time	433.88	41.88	395.97	47.65	430.19	42.05
Performance	n.a.	n.a.	n.a.	n.a.	90.84	7.48
Number of trials	201.06	71.37	16.63	10.94	217.69	69.51
Accuracy-instruction from Mattes et al. (2022) **(N = 61)**						
EEG amplitude	−1.93	3.10	−8.45	8.62	−2.31	3.02
EEG reliability	0.97	0.02	0.86	0.13	0.97	0.02
Response time	545.01	38.55	552.34	108.70	544.41	39.80
Performance	n.a.	n.a.	n.a.	n.a.	94.78	4.65
Number of trials	272.79	53.70	14.36	10.20	287.15	48.92
Speed-instruction from Mattes et al. (2022) **(N = 61)**						
EEG amplitude	−2.16	2.74	−7.95	4.32	−3.50	2.68
EEG reliability	0.84	0.09	0.77	0.13	0.85	0.08
Response time	462.63	56.36	341.45	63.00	430.89	57.63
Performance	n.a.	n.a.	n.a.	n.a.	73.27	8.94
Number of trials	84.10	40.90	28.38	8.75	112.48	42.17

Note. n.a. (not applicable). All statistics were calculated based on the trials included in the analysis. Note that the means and standard deviations were computed across individual subjects' averages, while the main analysis was performed on the single-trial data.

for the null effect, for the boundary separation it was inconclusive (BF = 1.10). In the no-instruction dataset, we observed very different pattern of results. The CRN effect on both the boundary separation and drift rate was robust and larger than the ERN amplitude effect (boundary separation: b = −0.03 for ERN, b = −0.05 for CRN; drift rate: b = −0.05 for ERN, b = −0.13 for CRN).

Unlike our dataset, we did not observe an effect of the Flanker condition in either of the datasets from Mattes et al. (2022). In particular, the no-instruction, speed, and accuracy datasets showed no association between condition type and either increased response caution or a slower drift rate.

Figs. 6 and 7 visualize the variability between participants in the effects of ERN and CRN amplitudes on the boundary separation and the drift rate for all four datasets.

Individual-level effects differ considerably across the four analyzed datasets. In both the current and speed-instruction datasets, some individuals exhibited an effect of ERN amplitude on the boundary separation, while others showed no effect. In the accuracy-instruction dataset, nearly all individuals demonstrated a reliable effect of ERN amplitude, whereas in the no-instruction dataset, almost all participants displayed the null effect. In contrast, the individual-level effects of CRN amplitude were more consistent across datasets. In the current, accuracy-instruction, and speed-instruction datasets, nearly all participants showed no effect of CRN amplitude. However, in the no-instruction dataset, a considerable number of participants exhibited an effect of CRN amplitude on the boundary separation.

The effects of ERN and CRN amplitude on the drift rate show similar patterns of consistency across datasets as those observed for the boundary separation. In both the current and accuracy-instruction datasets, we found effects of ERN amplitude of similar magnitude. In the speed-instruction dataset, most individuals exhibited an effect of ERN amplitude, though some showed a null effect. Additionally, the posterior distributions in the speed-instruction dataset were notably wider compared to the other datasets. In the no-instruction dataset, we consistently observed a null effect across all individuals. There was a consistent lack of CRN amplitude effect in the current, accuracy-instruction, and speed-instruction datasets. However, in the no-instruction dataset, the effect of CRN amplitude was stronger than that of ERN.

Overall, the effects of ERN and CRN from the preceding trial on DDM parameters in the current, accuracy-instruction, and speed-instruction datasets show moderate alignment. The findings from the speed-instruction dataset exhibit greater uncertainty in comparison to other datasets, making them mostly inconclusive. The no-instruction dataset, however, differs significantly from the current and accuracy-instruction datasets. It shows a lack of interaction effects between EEG amplitude and response type on drift rate, along with a stronger effect of CRN amplitude compared to ERN amplitude on both drift rate and boundary separation in subsequent trials. The boundary separation results display greater variability between datasets than the drift rate results. Furthermore, unlike our dataset, no effects of the Flanker condition were observed in the no-instruction, speed, or accuracy datasets.

Detailed results for the fixed and random effects in all three models, as well as the model convergence checks, are available on OSF.

3.5. Correlation analysis

Individual differences in the effects of ERN/CRN amplitudes on subsequent adaptation may arise from various sources, including noise, methodology-related factors, and trait-related individual differences. This study aims to identify potential moderators of the association between ERN/CRN amplitudes and indicators of cognitive adjustments derived from DDM model: boundary separation and drift rate.

Moderators included (1) the number of erroneous and correct trials used in the DDM analysis; (2) the number of errors committed during the task; (3) ERPs consistency calculated as dependability coefficient; and (4) trait-related individual differences, including anxiety, behavioral inhibition, intolerance of uncertainty (inhibitory and prospective dimensions), obsessive-compulsive symptoms, worry, and decisiveness (see the Exploratory correlation analysis section in the Method section). The results of the exploratory correlation analysis are shown in Table 6.

Table 5
Posterior means (*b*), standard deviations of the posterior distributions (*SD*), 95% Highest Posterior Density Intervals (*HDIs*), and Bayes Factors (*BF*) for the fixed effects on the boundary separation and drift rate, across the no-instruction, accuracy-instruction, and speed-instruction datasets.

Predictors	Boundary separation α				Drift rate δ			
	b	*SD*	*HDIs*	*BF*	*b*	*SD*	*HDIs*	*BF*
No-instruction (Base) from Mattes et al. (2022)								
(Intercept)	1.23	0.02	[1.19, 1.27]	n.a	2.72	0.1	[2.51, 2.91]	n.a
Condition type $\mu_{(c)}$	−0.02	0.01	[−0.04, 0.01]	0.06	−0.07	0.04	[−0.16, 0.01]	0.08
EEG amplitude $\mu_{(e)}$	−0.04	0.01	[−0.05, −0.03]	4.85×10^{49}	−0.09	0.02	[−0.13, −0.04]	46.93
Response type $\theta_{(r)}$	−0.06	0.01	[−0.08, −0.04]	8.40×10^{42}	0.16	0.04	[0.07, 0.24]	44.34
EEG amplitude × Response type $\mu_{(er)}$	−0.01	0.01	[−0.02, 0.01]	0.12	−0.04	0.03	[−0.09, 0.01]	0.16
EEG amplitude × Condition type $\theta_{(ec)}$	0.00	0.01	[−0.01, 0.01]	0.04	−0.01	0.02	[−0.05, 0.05]	0.05
Response type × Condition type $\theta_{(rc)}$	0.00	0.01	[−0.02, 0.02]	0.05	0.07	0.04	[−0.01, 0.16]	0.3
EEG amplitude × Response type × Condition type $\theta_{(erc)}$	0.00	0.01	[−0.02, 0.01]	0.04	−0.01	0.02	[−0.05, 0.04]	0.05
ERN amplitude $\mu_{(EN)}$	−0.03	0.01	[−0.05, −0.01]	5.70	−0.05	0.03	[−0.11, 0.02]	0.19
CRN amplitude $\mu_{(CN)}$	−0.05	0.01	[−0.07, −0.03]	4.39×10^{35}	−0.13	0.04	[−0.20, −0.06]	59.19
Intercept variability σ	0.18	0.02	[0.15, 0.21]	n.a	0.87	0.07	[0.74, 1.01]	n.a
Condition type variability $\sigma_{(c)}$	0.02	0.01	[0.01, 0.03]	n.a	0.03	0.02	[0.01, 0.07]	n.a
EEG amplitude variability $\sigma_{(e)}$	0.02	0.01	[0.01, 0.04]	n.a	0.02	0.02	[0.01, 0.06]	n.a
EEG amplitude × Response type variability $\sigma_{(er)}$	0.02	0.01	[0.01, 0.04]	n.a	0.03	0.02	[0.01, 0.07]	n.a
Accuracy-instruction from Mattes et al. (2022)								
(Intercept)	1.59	0.05	[1.49, 1.68]	n.a	3.30	0.14	[3.02, 3.57]	n.a
Condition type $\mu_{(c)}$	0.04	0.02	[−0.01, 0.07]	0.01	0.03	0.05	[−0.08, 0.14]	0.04
EEG amplitude $\mu_{(e)}$	−0.06	0.01	[−0.08, −0.04]	5.25×10^{13}	−0.14	0.04	[−0.21, −0.07]	78.19
Response type $\theta_{(r)}$	−0.06	0.02	[−0.09, −0.02]	5.96	0.08	0.06	[−0.03, 0.19]	0.33
EEG amplitude × Response type $\mu_{(er)}$	0.04	0.01	[0.01, 0.06]	5.71	0.12	0.04	[0.05, 0.20]	13.69
EEG amplitude × Condition type $\theta_{(ec)}$	0.01	0.01	[−0.01, 0.03]	0.09	0.02	0.04	[−0.04, 0.10]	0.09
Response type × Condition type $\theta_{(rc)}$	−0.04	0.02	[−0.07, - 0.01]	0.75	−0.01	0.05	[−0.12, 0.09]	0.12
EEG amplitude × Response type × Condition type $\theta_{(erc)}$	0.01	0.01	[−0.01, 0.03]	0.08	0.05	0.04	[−0.02, 0.12]	0.19
ERN amplitude $\mu_{(EN)}$	−0.1	0.02	[−0.15, −0.06]	146.62	−0.26	0.07	[−0.39, −0.12]	109.04
CRN amplitude $\mu_{(CN)}$	−0.02	0.01	[−0.04, −0.01]	1.10	−0.01	0.03	[−0.06, 0.04]	0.06
Intercept variability σ	0.34	0.04	[0.28, 0.41]	n.a	0.99	0.10	[0.82, 1.19]	n.a
Condition type variability $\sigma_{(c)}$	0.02	0.01	[0.01, 0.03]	n.a	0.03	0.02	[0.01, 0.07]	n.a
EEG amplitude variability $\sigma_{(e)}$	0.02	0.01	[0.01, 0.03]	n.a	0.03	0.02	[0.01, 0.07]	n.a
EEG amplitude × Response type variability $\sigma_{(er)}$	0.01	0.01	[0.01, 0.03]	n.a	0.03	0.02	[0.01, 0.08]	n.a
Speed-instruction from Mattes et al. (2022)								
(Intercept)	1.25	0.02	[1.21, 1.30]	n.a	1.04	0.07	[0.90, 1.17]	n.a
Condition type $\mu_{(c)}$	0.00	0.01	[−0.02, 0.02]	0.01	0.02	0.03	[−0.04, 0.08]	0.02
EEG amplitude $\mu_{(e)}$	−0.03	0.01	[−0.05, −0.01]	4.49	−0.02	0.03	[−0.08, 0.03]	0.08
Response type $\theta_{(r)}$	0.04	0.01	[0.03, 0.06]	5.13×10^{15}	0.16	0.03	[0.11, 0.22]	1.94×10^{26}
EEG amplitude × Response type $\mu_{(er)}$	0.04	0.01	[0.02, 0.05]	2.52×10^{3}	0.06	0.03	[0.01, 0.11]	0.41
EEG amplitude × Condition type $\theta_{(ec)}$	0.00	0.01	[−0.03, 0.01]	0.09	−0.02	0.03	[−0.07, 0.03]	0.08
Response type × Condition type $\theta_{(rc)}$	0.00	0.01	[−0.01, 0.02]	0.04	0.04	0.03	[−0.02, 0.10]	0.17
EEG amplitude × Response type × Condition type $\theta_{(erc)}$	0.00	0.01	[−0.02, 0.01]	0.04	0.00	0.03	[−0.05, 0.05]	0.05
ERN amplitude $\mu_{(EN)}$	−0.06	0.01	[−0.09, −0.04]	2.38×10^{11}	−0.08	0.05	[−0.17, 0.01]	0.43
CRN amplitude $\mu_{(CN)}$	0.01	0.01	[−0.01, 0.03]	0.07	0.03	0.03	[−0.03, 0.09]	0.11
Intercept variability σ	0.16	0.02	[0.12, 0.19]	n.a	0.49	0.05	[0.39, 0.59]	n.a
Condition type variability $\sigma_{(c)}$	0.03	0.01	[0.01, 0.05]	n.a	0.05	0.03	[0.01, 0.10]	n.a
EEG amplitude variability $\sigma_{(e)}$	0.03	0.01	[0.01, 0.05]	n.a	0.04	0.03	[0.01, 0.09]	n.a
EEG amplitude × Response type variability $\sigma_{(er)}$	0.02	0.01	[0.01, 0.04]	n.a	0.05	0.03	[0.01, 0.11]	n.a

Note. c, *e*, and *r* denote condition, ERN/CRN amplitude, and response type, respectively. Detailed descriptions of the predictors are provided in Table 1. Bayes Factor was calculated as Savage–Dickey ratio. *BF* above 5 indicates an evidence for an effect, *BF* between 5 and . 20 suggests insufficient evidence to determine the presence of the effect, and *BF* lower than . 20 indicates strong evidence for the null effect. *BF* $>10^{308}$ indicates a BF value exceeding the double-precision floating-point limit. Note that EEG, ERN, and CRN amplitudes, as well as response type, come from the preceding trial.

Among potential moderators, only the effect of ERN amplitude on boundary separation was significantly associated with methodology-related factors and behavioral results. Specifically, the effect of ERN amplitude on boundary separation correlated with the number of error trials (BF = 23.16), correct trials (BF = 78.46) included in the analysis, and the total number of committed errors (BF = 182.90). In contrast, Bayes Factors for the effect of CRN on boundary separation were inconclusive, with BF = 2.67, 1.82, and 3.36 for the number of error trials, correct trials, and committed errors, respectively. Neither the ERN nor CRN effects on boundary separation were associated with self-report traits, with Bayes Factors providing substantial evidence for the null in most cases.

Conversely, Bayes Factors provided evidence for the absence of associations between the effects of ERN and CRN amplitudes on drift rate and both the number of error trials and ERP consistency, suggesting that these effects are less dependent on methodology. Notably, the effect of ERN amplitude on drift rate was associated with behavioral inhibition (BF = 55.88), and showed marginal associations with the inhibitory dimension of intolerance of uncertainty (BF = 2.71) and worry (BF = 3.21), although these latter findings fall short of strong evidential support. The effect of CRN amplitude on drift rate was not associated with any moderators, and Bayes Factors consistently supported the null hypothesis.

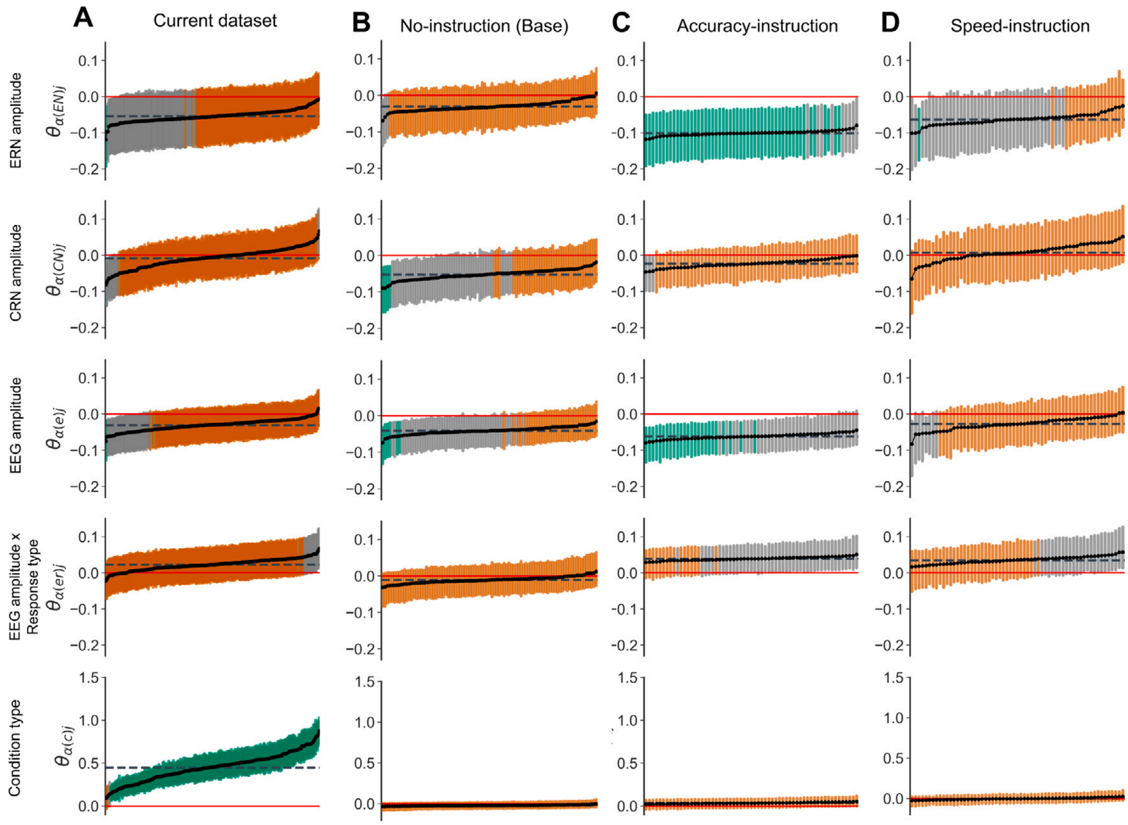

Fig. 6. Individual ERN/CRN amplitudes effects ($\theta_{(e)j}$) and their interactions with response type ($\theta_{(er)j}$), along with the effects of ERN ($\theta_{EN(j)}$), CRN ($\theta_{CN(j)}$), and condition type ($\theta_{(c)j}$) on the boundary separation α for the current (panel A), no-instruction (panel B), accuracy-instruction (panel C), and speed-instruction (panel D) datasets. The dashed horizontal line denotes the population-level posterior means of the effects. Credible intervals (CrIs) are colored as follows: orange for Bayes Factors less than 0.33, gray for Bayes Factors between 0.33 and 3, and green for Bayes Factors greater than 3. Note that EEG, ERN, and CRN amplitudes, as well as response type, come from the preceding trial.

4. Discussion

Using neuro-cognitive diffusion modeling, Mattes et al. (2022) recently demonstrated the association between neural correlates of response monitoring, specifically ERN and CRN, and indicators of subsequent cognitive adjustment. Mattes and colleagues' study provided novel insights into the mechanisms underlying post-response adaptation, showing that more pronounced ERN/CRN amplitudes were specifically associated with increased attention to stimuli and greater response caution in subsequent trials. To further our understanding of the cognitive processes involved in adaptive behavior and the role played by neural correlates of response monitoring in these processes, we reanalyzed the datasets from Mattes et al. (2022) and conducted analyses on a new large-scale dataset. Unlike previous studies, which primarily focused on group-level effects, our study is the first to systematically examine between-participant variance in these associations. This approach allowed us to evaluate whether the relationship between response-related brain activity and subsequent cognitive adaptation is robust across individuals and datasets. By doing so, we established the extent to which ERN amplitudes reliably predict subsequent adaptation at the individual level, beyond general group-level trends, and whether the strength of this prediction depends on external factors, such as task instructions, or internal factors, such as individual traits.

4.1. Population-level effects

At the population level, our findings from the current, no -instruction, speed-instruction, and accuracy-instruction datasets were consistent with those reported by Mattes et al. (see Supplementary Table 4 by Mattes et al., 2022). We observed a stable association between the EEG signal and boundary separation in the subsequent trial across all datasets. This relationship was primarily driven by the EEG signal following error commission, i.e., the ERN, in all datasets except for the no-instruction dataset. These results indicate a robust population-level link between the magnitude of response-related brain activity, particularly post-error, and an increase in response caution. Since increasing the response caution helps prevent future errors by allowing more evidence to accumulate before making a decision, our results support the association between the ERN and adaptive processes. Importantly, this pattern did not generally hold for the CRN, suggesting functional differences between the ERN and CRN. We also observed a consistent association between ERN amplitude and drift rate, and a lack of similar association with CRN amplitude, across the current, accuracy-instruction, and speed-instruction datasets. This pattern suggests a robust population-level relationship between response-related brain activity and increased attention to task-relevant features, particularly following errors. These findings align with the idea that errors prompt a shift toward enhanced stimulus-focused attention (Danielmeier and Ullsperger, 2011).

4.2. Individual-level effects

Despite a robust population-level effect of ERN amplitude on the boundary separation, we observed substantial variability in the individual-level effects, with distinct patterns across different datasets. We hypothesize that this variability may be driven by individual differences in managing speed–accuracy trade-off. Speed–accuracy manipulations are known to influence response caution, potentially limiting full adjustment under speed-focused conditions (Hedge et al., 2019). In such cases, the relationship between brain activity and subsequent changes in response caution may not be fully evident, particularly not uniformly across individuals. Although our paradigm emphasized

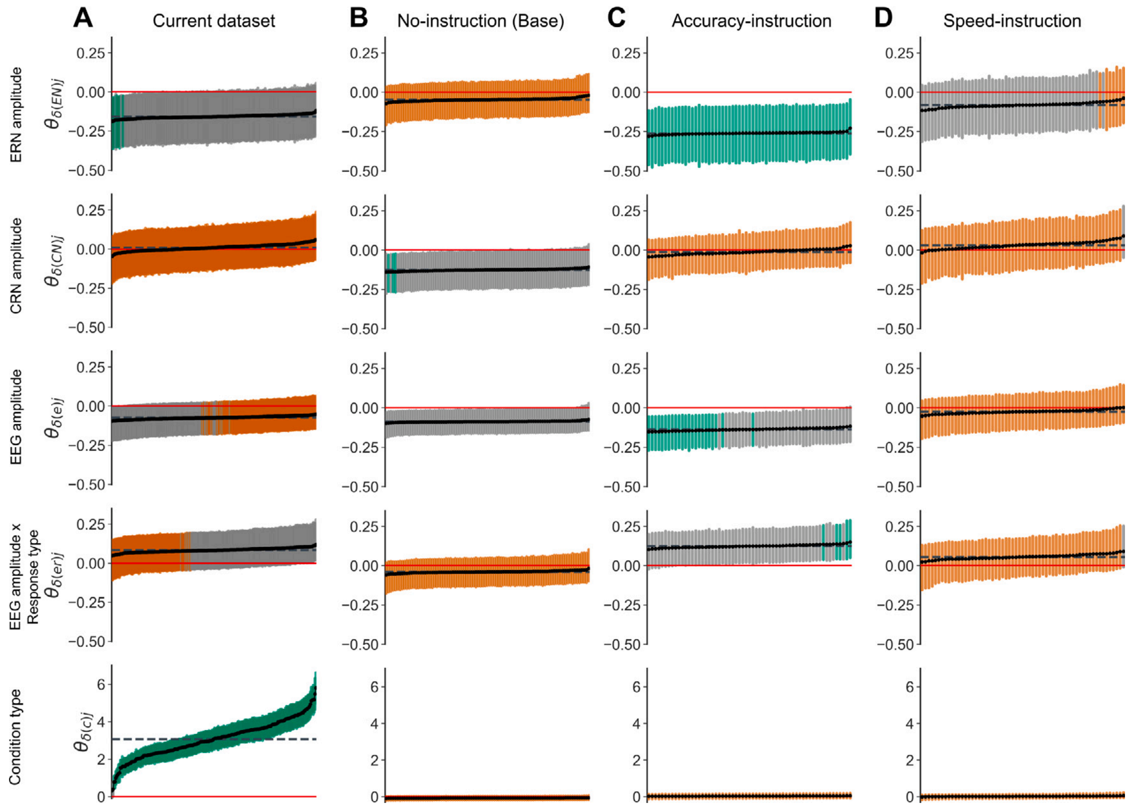

Fig. 7. Individual ERN/CRN amplitudes effects ($\theta_{(e)j}$) and their interactions with response type ($\theta_{(er)j}$), along with the effects of ERN ($\theta_{EN(j)}$), CRN ($\theta_{CN(j)}$), and condition type ($\theta_{(c)j}$) on the drift rate δ for the current (panel A), no-instruction (panel B), accuracy-instruction (panel C), and speed-instruction (panel D) datasets. The dashed horizontal line denotes the population-level posterior means of the effects. Credible intervals (CrIs) are colored as follows: orange for Bayes Factors less than 0.33, gray for Bayes Factors between 0.33 and 3, and green for Bayes Factors greater than 3. Note that EEG, ERN, and CRN amplitudes, as well as response type, come from the preceding trial.

Table 6
Bayesian correlation matrix of performance-related and self-reported motivational and affective factors, and the effects of ERN and CRN amplitudes on boundary separation and drift rate. Means (*M*) and standard deviations (*SD*) of the variables are also included.

	M	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. CRN amplitude effect on boundary separation	−.01	.03	–	*−.01*	**.21**	−.05	.08	−.07	.16	−.15	.17	*.03*	.05	*.00*	*−.01*	*.01*	*−.04*	*.00*
2. ERN amplitude effect on boundary separation	−.05	.02		–	*−.01*	**−.25**	.06	−.14	**.21**	**−.24**	**.25**	.09	.09	.05	*−.01*	*−.01*	*−.01*	.05
3. CRN amplitude effect on drift rate	.01	.02			–	**−.39**	−.08	*.04*	*−.05*	.05	−.08	*.02*	−.09	*.03*	*−.03*	*.04*	*−.05*	*.02*
4. ERN amplitude on drift rate	−.16	.01				–	*.00*	.05	*−.04*	.08	−.05	.10	.16	.14	**.23**	.05	.17	−.16
5. ERN amplitude consistency	.92	.05					–	.13	**.61**	**−.44**	**.52**	−.13	*−.04*	−.13	*−.02*	*−.02*	−.09	*.02*
6. CRN amplitude consistency	.98	.01						–	**−.37**	**.51**	**−.47**	*−.03*	.06	*−.03*	−.06	−.05	*.01*	*.03*
7. Number of error trials in the analysis	24.95	8.26							–	**−.86**	**.91**	−.08	−.06	−.10	*−.02*	*.03*	−.08	.02
8. Num correct trials in the analysis	202.17	37.46								–	**−.95**	.08	.09	.10	.13	*−.02*	.14	−.09
9. Number of errors committed during the task	34.42	15.68									–	−.09	−.09	−.12	−.09	*.01*	−.13	.06
10. Prospective intolerance of uncertainty	3.15	.77										–	**.47**	**.35**	**.30**	**.24**	**.34**	**−.19**
11. Inhibitory intolerance of uncertainty	2.43	.90											–	**.60**	**.40**	**.30**	**.48**	**−.32**
12. Anxiety trait	2.28	.46												–	**.60**	**.38**	**.75**	**−.38**
13. Behavioral inhibition	3.00	.51													–	**.26**	**.71**	**−.43**
14. Obsessive-compulsive symptoms severity	2.07	.56														–	**.38**	**−.35**
15. Worry	3.26	.88															–	**−.39**
16. Decisiveness	3.24	.99																–

Note. The self-report measures were obtained using Polish adaptations or translations of standardized questionnaires, developed through a forward–backward translation protocol. ERN = error-related negativity; ERN amplitude = peak ERN amplitude at FCz; CRN = correct-related negativity; CRN amplitude = peak CRN amplitude at FCz. Bolded Bayes Factors (BF >5) indicate moderate to strong evidence for the alternative hypothesis, while italicized Bayes Factors (BF <0.2) indicate moderate to strong evidence for the null hypothesis. Full Bayes Factors are available in the Supplementary Materials: https://osf.io/chgan/files/osfstorage. Self-report measures were calculated as total scores following the original scoring instructions, and normalized by the number of items.

both speed and accuracy, the observed short response times suggest that participants prioritized speed, thus influencing response caution adjustment. This interpretation is further supported by the similarity in results between the current and the speed-instruction datasets, and the stronger association between ERN amplitude and response caution observed in the accuracy-instruction dataset, where adaptive adjustments were not constrained by task conditions.

Further, the exploratory correlation analysis revealed that the effect size of ERN amplitudes on boundary separation in the current dataset was positively correlated with the number of committed errors and number of erroneous trial used in the analysis, indicating that stronger effects were observed in participants with less errors. Notably, this association was not observed for the effect of CRN amplitude on boundary separation.

Performance may moderate the effect of ERN amplitude on boundary separation for three very different reasons. First, individuals with better performance can maintain a more effective speed–accuracy trade-off, and their implementation of response caution is less influenced by speed pressure. In such cases, the ERN may signal a need for an increase in response caution, which may then be implemented, or not, depending on external circumstances. This hypothesis might also explain the differences observed between datasets. Second, studies have shown that ERN amplitudes tend to diminish as the number of errors increases, implying that a habituation process reduces the brain's response to errors over time (Boksem et al., 2006; Fischer et al., 2017; Senderecka et al., 2018). This gradual attenuation across the duration of the task may interfere with accurately estimating the impact of ERN amplitude on DDM parameters. Third, in a recent study, Lüken et al. (2023) showed that high performance, marked by a low number of errors, can introduce bias in the DDM model and result in an overestimation of effects at low error rates.

Although the performance metrics for the no-instruction dataset were similar to those of the current dataset, the association between ERN amplitude and boundary separation was not present for most individuals. Notably, in this dataset, CRN amplitude was associated with boundary separation in over half of the participants. The origins of this reversed pattern remain unclear.

Overall, our study demonstrated that while a relationship between ERN amplitude and boundary separation exists, it is complex and context-dependent, as illustrated in Fig. 1, panel B. Factors such as task design are strong moderators that can change the direction of the effect of ERN amplitude on post-error adjustments in response caution. Moreover, this considerable variability in individual-level effects suggests that population-level estimates may be misleading, as they average across individuals who do show the effect and those who do not.

Variability in individual-level effects of ERN amplitude on drift rate was smaller than for boundary separation but still varied between datasets. In the current and accuracy-instruction datasets all participants showed the effect consistently. In contrast, all participants in the speed-instruction dataset exhibited smaller, more uncertain effect, and individual from the no-instruction dataset showed no effects. These findings suggest that the relationship between ERN amplitude and increased attention to stimuli may be influenced by some task design-related factors. Nonetheless, the high consistency of the effect in the current and accuracy-instruction datasets also supports the idea that ERN amplitude may serve as a relatively reliable indicator of adaptive processes aimed at enhancing stimulus-focused attention. The distinct pattern in the no-instruction dataset, where the CRN showed an effect while the ERN did not, warrants further investigation. Understanding why the CRN, rather than the ERN, predicted adaptation in this specific context could offer valuable insights into the mechanisms of cognitive control and clarify the distinct contributions of response-related brain activity.

Further, correlation analysis revealed that anxiety-related individual differences, specifically behavioral inhibition, moderate the strength of the relationship between ERN amplitude and drift rate. Specifically, higher levels of behavioral inhibition were linked to a weaker effect of ERN on drift rate. This suggests that behavioral inhibition is a weak moderator of the relationship between ERN amplitude and post-error adjustments in drift rate. While this trait has often been linked to ERN amplitude, our analysis is the first to show that it also shapes how ERN amplitude relates to subsequent cognitive adaptation.

These findings support the connection between negative affect and cognitive control, as outlined in the Adaptive Control Hypothesis (TACH). Proposed by Shackman et al. (2011), TACH emphasizes the interplay between emotional and cognitive processes in adaptive control, suggesting that negative affect, pain, and cognitive control share common neural mechanisms due to their overlapping roles in managing aversive and uncertain situations (Cavanagh and Shackman, 2015). A meta-analysis by Shackman et al. (2011) provides strong evidence for this functional overlap, demonstrating that both these processes consistently engage the anterior midcingulate cortex (aMCC) in tasks such as go/no-go or Flanker. Within this framework, the aMCC serves as a central hub, integrating information about pain, threat, and punishment to regulate goal-directed behavior, fear expression, and attentional control. Notably, both transient emotional states (e.g., pain or threat) and stable anxiety-related traits have been shown to influence cognitive control mechanisms within the MCC (Cavanagh and Shackman, 2015; Grupe and Nitschke, 2013). Our results provide evidence that elevated levels of anxiety-related traits can indeed disrupt cognitive control, particularly by interfering with adaptive processes that enhance stimulus-focused attention.

Importantly, TACH primarily focuses on frontal midline theta oscillations. A growing body of research points to frontal midline theta oscillations as a key component of cognitive control (Cavanagh and Frank, 2014; Cavanagh and Shackman, 2015; Beatty et al., 2021). While our study focused on the ERN, prior work suggests that theta activity, from which the ERN is derived, may reflect a distinct neurocognitive process more directly involved in implementing control. In this view, frontal midline theta supports the execution of adaptive responses, whereas the ERN reflects the affective aspect of action monitoring or signals the need for adjustment. Whether the observed ERN's association with behavioral adjustment is mediated by theta-related control processes, or is specific to ERN, remains to be clarified.

4.3. Limitations and further directions

While our study provides valuable insights into the relationship between ERN amplitude and cognitive adaptation, several limitations warrant consideration.

First, we did not model the effects of EEG amplitudes, response types, and condition types on non-decision time. Given that our model already included a large number of parameters, we focused on DDM parameters that were central to our hypotheses. Nevertheless, although we did not specifically model this relationship, our estimates of effects on boundary separation and drift rate align with those reported by Mattes et al. (2022), who modeled effects on non-decision time. This suggests that the inclusion of non-decision time in the analysis would likely not have changed our results significantly. However, future studies should explore the role of non-decision time in adaptive behavior and response monitoring more thoroughly.

Second, the relatively small number of errors in the data constrained estimation precision, leading to large individual credible intervals. To improve precision, a larger number of trials may be necessary. This could enhance estimation accuracy and yield more diagnostic results, including improved correlation analyses results.

Third, it is important to recognize that even correct trials can involve partial errors, which are known to elicit ERN-like activity (see Ficarella et al., 2019; Vidal et al., 2022). These subtle response conflicts, often overlooked, may contribute to variability in the CRN and

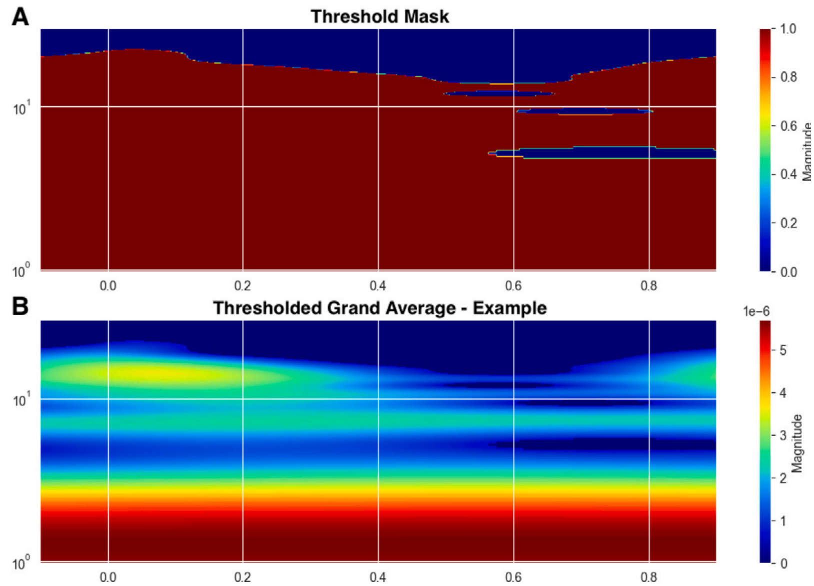

Fig. A.8. Estimated wavelet filter (panel A) at FCz electrode. The wavelet filter covered frequencies from 1 to 30 Hz in 200 steps, spaced evenly on a logarithmic scale, with a cutoff of 0.45. Example of thresholded grand average (panel B) at FCz electrode.

complicate interpretations of its role in performance monitoring (see Maruo et al., 2017; Maruo and Masaki, 2022). As such, the presence of partial errors may influence observed CRN effects, highlighting the need for future research to more systematically account for them. Additionally, it has been shown that correction behavior influences post-error adjustments (Beatty et al., 2021; Kalfaoglu et al., 2018; Crump and Logan, 2013). Future research could benefit from including measures of both overt and subthreshold corrections and considering how task instructions related to error correction may shape these adjustments.

Fourth, the peak amplitude measure is known to be highly sensitive to single-trial peak latencies (Mattes et al., 2023). Thus, mean ERN amplitude may be a more stable metric, as it is less influenced by external factors such as peak latency variability.

Finally, while our study identifies several potential moderators, such as task conditions and internal factors like anxiety traits, it is important to explore additional moderators that may influence the relationship between response monitoring and behavioral adaptation. These might include factors such as individual differences in executive control, motivation, or stress reactivity are particularly promising candidates, as these factors are known to influence ERN amplitude and are associated with overlapping neural circuits involved in cognitive control and emotion regulation (Grupe and Nitschke, 2013). Identifying and testing these potential moderators will be key to refining our understanding of the complex interplay between cognitive control and emotional processes in adaptive behavior. Therefore, future research should aim to expand on this work by investigating other moderators and their impact on the adaptive processes we have studied.

5. Conclusions

To better understand the cognitive mechanisms underlying adaptive behavior and the role of neural correlates of response monitoring, we investigated whether the association between neural response monitoring and subsequent cognitive adjustments is consistent across individuals. Our findings reveal a complex picture. At the population level, more pronounced ERN amplitudes were associated with increased response caution, reflected in boundary separation, and enhanced attention to task-relevant stimuli, as indicated by drift rate. However, these effects did not align with individual-level patterns. The relationship between ERN amplitude and response caution showed substantial variability and was moderated by task characteristics such as speed–accuracy trade-offs and performance, while remaining largely unaffected by anxiety-related traits. In contrast, the link between ERN amplitude and drift rate was more consistent across individuals and showed modest influence from behavioral inhibition. Assuming that boundary separation reflects response caution and drift rate reflects attentional engagement, our study is the first to demonstrate that ERN amplitude is not equally predictive of these distinct cognitive adjustments, which becomes apparent only through individual-level analysis. Our findings underscore the importance of moving beyond population-level averages to account for individual variability when interpreting the functional significance of neural markers like the ERN in adaptive behavior.

CRediT authorship contribution statement

Anna Grabowska: Visualization, Investigation, Conceptualization, Methodology, Formal analysis, Writing – original draft, Software, Data curation. **Filip Sondej:** Software, Data curation, Writing – review & editing. **Julia Haaf:** Funding acquisition, Supervision, Conceptualization, Writing – review & editing, Methodology. **Michael D. Nunez:** Writing – review & editing, Conceptualization, Supervision, Methodology, Software. **Magdalena Senderecka:** Methodology, Supervision, Project administration, Conceptualization, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

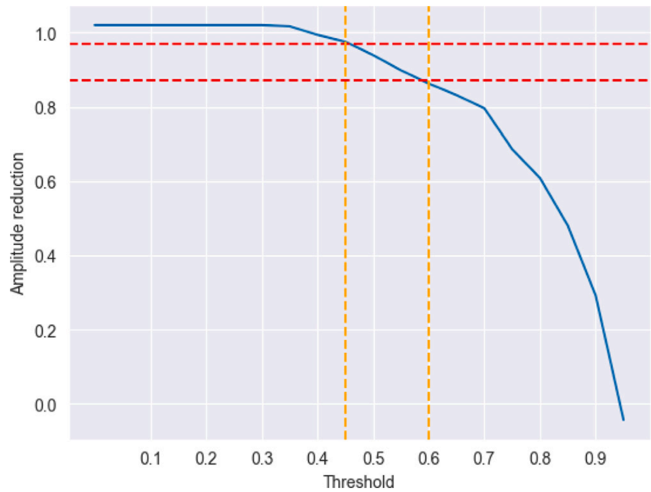

Fig. A.9. Cutoff selection. Before selecting the cutoff, we estimated the relationship between cutoffs and amplitude reduction. The chosen cutoff of 0.45 is marked by an orange dashed line, and the amplitude reduction at this cutoff is marked by a red dashed line.

Acknowledgments

We thank Maarten Marsman for advice on calculating Bayes Factors for hierarchical models, and André Mattes and his collaborators for providing additional data in support of the analysis. We gratefully acknowledge the help of students and collaborators from Jagiellonian University with the data recording. We are also grateful to the participants who volunteered to take part in the study. We gratefully acknowledge Polish high-performance computing infrastructure PLGrid (HPC Center: ACK Cyfronet AGH) for providing computer facilities and support within computational grant no. PLG/2025/018163. Article Processing Charges were funded by a grant from the Strategic Program Excellence Initiative at the Jagiellonian University.

Funding

This study was supported by a Sonata Bis 10 grant (2020/38/E/HS6/00490) from the National Science Centre of Poland awarded to Magdalena Senderecka. Julia Haaf was supported by a Veni Grant from the Dutch Research Council (VI.Veni.201G.019).

Appendix A. Pre-processing

To enhance the signal-to-noise ratio of single-trial ERPs (as estimates of CRN/ERN on single-trials), we used wavelet filtering (see Quiroga and Garcia, 2003; see also Hu et al., 2010; Mattes et al., 2022). Wavelet filters help remove signals that are not systematically linked to processes of interest by zeroing out wavelet coefficients with low energy, which presumably contain little relevant information. The proportion of wavelet coefficients to be zeroed is determined by the threshold parameter. Here, we built a wavelet filter based on all error trials for FCz and Cz separately. The wavelet filter covered frequencies from 1 to 30 Hz in 200 steps, spaced evenly on a logarithmic scale (Torrence and Compo, 1998; Thakur et al., 2013; Lilly, 2017), and had a cutoff of 0.45. Fig. A.8 shows the created wavelet filter (panel A) and signal grand average with applied filter (panel B) at electrode FCz.

When choosing the threshold value, we followed the recommendations of Hu et al. (2010), such that an ideal cutoff is just before the curve's inflection point, where the amplitude begins to drop significantly with increasing thresholds due to the increasing removal of the true signal (see Figure 2 by Hu et al., 2010). This approach ensures that the maximum amount of noise is eliminated from the data while preserving as much of the original amplitude as possible. In our wavelet filter created with a logarithmic scale, the threshold of 0.45 resulted in an amplitude reduction of approximately 5% from the base reduction at a threshold of 0 (see Fig. A.9), similar to Mattes et al. (2022). The wavelet-filtered single-trial CRN/ERNs were then defined as peak amplitudes in the interval from the response onset until 100 ms after the response onset at FCz electrode.

Table A.7 summarize the differences between pre-processing of the current dataset and the datasets from Mattes et al. (2022).

Appendix B. Specification of non-cognitive model

B.1. Response times

To estimate the effect of ERN and CRN on subsequent behavioral adaptation, we regressed the RTs on the EEG amplitudes and response type (error vs. correct) from the preceding trial using multilevel linear regression. The definition of the response time model was as follow,

$$\begin{aligned} rt_{ij} \sim Normal(&\alpha_j + \theta_{(c)j} * \mathrm{c}_i + \theta_{(e)j} * \mathrm{e}_i + \theta_{(er)j} * \mathrm{e}_i * \mathrm{r}_i + \\ &+ \theta_{(r)} * \mathrm{r}_i + \theta_{(ec)} * \mathrm{e}_i * \mathrm{c}_i + \theta_{(rc)} * \mathrm{r}_i * \mathrm{c}_i + \theta_{(erc)} * \mathrm{e}_i * \mathrm{r}_i * \mathrm{c}_i, \sigma) \end{aligned}$$

where i indicates trial number and j indicates the individual to whom the trial belongs; c_i is a condition type at trial i; r_i is a response type preceding trial i; e_i is a EEG response-related amplitude preceding trial i. θ denotes effects listed in Table 1.

For random intercept α_j and slopes θ_j, fixed effects, and variability, we used the following priors:

$$\begin{aligned} \begin{bmatrix} \alpha_j \\ \theta_{(c)j} \\ \theta_{(e)j} \\ \theta_{(er)j} \end{bmatrix} &\sim \text{Normal}\left(\begin{bmatrix} \mu_\alpha, \sigma_\alpha \\ \mu_{\theta_{(c)}}, \sigma_{\theta_{(c)}} \\ \mu_{\theta_{(e)}}, \sigma_{\theta_{(e)}} \\ \mu_{\theta_{(er)}}, \sigma_{\theta_{(er)}} \end{bmatrix} \right) \\ \mu_\alpha &\sim \text{Normal}(.3, .5)[.1, .7] \\ \mu_{\theta_{(c)}}, \mu_{\theta_{(e)}}, \mu_{\theta_{(er)}}, \theta_{(r)}, \theta_{(ec)}, \theta_{(rc)}, \theta_{(erc)} &\sim \text{Normal}(0, .2) \\ \sigma, \sigma_{\theta(e)}, \sigma_{\theta(er)} &\sim \text{Normal}_+(0, .5) \\ \sigma_\alpha, \sigma_{\theta_{(c)}} &\sim \Gamma(.2, 1) \end{aligned} \tag{B.1}$$

Table A.7
Electroencephalography data pre-processing steps used in the current dataset (N=225) and those from Mattes et al. (2022), along with justifications for any modifications applied.

	Current dataset	Mattes et al. (2022)	Justification
1. Electrodes	64 active Ag/AgCl electrodes (BioSemi) and a BioSemi ActiveTwo DC amplifier	61 active Ag/AgCl electrodes (Brain Products) and a BrainAmp DC amplifier	–
2. Sampling rate	256 Hz	500 Hz	–
3. Reference	Average of the left and right mastoid electrodes	The left mastoid electrode	–
4. DC detrend correction	yes	yes	–
5. Epochs (around response)	from −100 to 900 ms	from −100 to 900 ms	–
6. Baseline correction	from −100 to 0 ms	from −100 to 0 ms	–
7. Ocular correction	Gratton et al. (1983) procedure	Gratton et al. (1983) procedure	–
8. Re-baselining	from −100 to 0 ms	from −100 to 0 ms	–
9. Artifact rejection	Epochs in which the EEG signal at the FCz or Cz site was greater than ± 150 μV were removed.	Epochs in which the EEG signal at the FCz or Cz site was greater than ± 150 μV were removed.	–
10. Single trial estimates	10.1. Denoising data with a wavelet filter (1 to 30 Hz in 200 steps spaced evenly on a log scale) for FCz and Cz electrodes separately. Threshold for wavelet filter at 0.45 (peak amplitude reduction ∼0.95)	10.1. Denoising data with a wavelet filter (1 to 30 Hz in steps of 0.3 Hz) for FCz and Cz electrodes separately. Threshold for wavelet filter at 0.85 (peak amplitude reduction ∼0.85)	For wavelet filters logarithmically spaced scales are recommended. Different threshold to maintain similar amplitude peak reduction from the plateau (5% from the base amplitude reduction when threshold = 0 is applied)
	10.2. Epochs: from −100 to 500 ms around response	10.2. Epochs: from −100 to 500 ms around response	–
	10.3. Additional operations: No	10.3. Additional operations: Multiple linear regression with distortion parameter by Hu et al. (2010)	The procedure changed the shape of single trial waveforms significantly. Because single trials after wavelet filtering looked reasonable, we decided to skip this step.
	10.4. Amplitude estimation: Peak amplitude in the interval from 0 to 100 ms after the response onset.	10.4. Amplitude estimation: Peak amplitude in the interval from 0 to 200 ms after the response onset.	The ERN/CRN peak should not be visible around 200 ms after response — thus, to avoid mistakenly choosing wrong peak (especially for CRN) we stuck to the classic 0–100 ms interval for peak search.
11. Trial selection	11.1. CCXP pattern	11.1. CCXP pattern	–
	11.2. RT criterion: remove P trials with a log-transformed RT above/below the mean of the log-transformed RT ±three standard deviations in a population-wise manner	11.2. RT criterion: remove P trials with a log-transformed RT above/below the mean of the log-transformed RT ±three standard deviations in a participant-wise manner	In our dataset some of the reaction times were very fast, making participant-level thresholds very wide. In our case, this kind of threshold is stronger (in the Mattes et al. (2022) datasets it is more conservative). Thus, we believe that population-level threshold is better suited for our data.

B.2. Accuracy

To estimate the effect of ERN and CRN on subsequent behavioral adaptation we regressed the accuracies on the EEG amplitudes and response type (error vs. correct) from the preceding trial. The definition of accuracy model was as follow,

$$\begin{aligned} correct_{(ij)} \sim Bernoulli(logistic(\alpha_j \\ &+ \theta_{(c)j} * \mathrm{c}_i \\ &+ \theta_{(e)j} * \mathrm{e}_i \\ &+ \theta_{(er)j} * \mathrm{e}_i * \mathrm{r}_i \\ &+ \theta_{(r)} * \mathrm{r}_i \\ &+ \theta_{(ec)} * \mathrm{e}_i * \mathrm{c}_i \\ &+ \theta_{(rc)} * \mathrm{r}_i * \mathrm{c}_i \\ &+ \theta_{(erc)} * \mathrm{e}_i * \mathrm{r}_i * \mathrm{c}_i)) \end{aligned} \tag{B.2}$$

where i indicates trial number and j indicates the individual to whom the trial belongs; c_i is a condition type at trial i; r_i is a response type preceding trial i; e_i is a EEG response-related amplitude preceding trial i. θ denotes effects listed in Table 1.

For random intercept α_j and slopes θ_j, fixed effects, and variability, we used the following priors:

$$\begin{aligned} \begin{bmatrix} \alpha_j \\ \theta_{(c)j} \\ \theta_{(e)j} \\ \theta_{(er)j} \end{bmatrix} &\sim \text{Normal}\left(\begin{bmatrix} \mu_\alpha, \sigma_\alpha \\ \mu_{\theta_{(c)}}, \sigma_{\theta_{(c)}} \\ \mu_{\theta_{(e)}}, \sigma_{\theta_{(e)}} \\ \mu_{\theta_{(er)}}, \sigma_{\theta_{(er)}} \end{bmatrix} \right) \\ \mu_\alpha &\sim \text{Normal}(0, 1.5) \\ \mu_{\theta_{(c)}}, \mu_{\theta_{(e)}}, \mu_{\theta_{(er)}}, \theta_{(r)}, \theta_{(ec)}, \theta_{(rc)}, \theta_{(erc)} &\sim \text{Normal}(0, .2) \\ \sigma_{\theta(e)}, \sigma_{\theta(er)} &\sim \text{Normal}_+(0, .5) \\ \sigma_\alpha, \sigma_{\theta_{(c)}} &\sim \Gamma(1, 1) \end{aligned} \tag{B.3}$$

Appendix C. Specification of DDM model

Let Y_{ijk} denote the response vector from the Flanker task, which includes both the response correctness and the response time (X_{ij}, T_{ij}), recorded for the ith trial, $i = 1, \ldots, I$, of the kth condition, $k = 1, \ldots, K$, of the jth participant, $j = 1, \ldots, J$. The observed bivariate data Y_{ijk} is assumed to follow a Wiener distribution, which is described as:

$$Y_{ijk} \sim \text{Wiener}(\alpha_{ijk}, \beta_{ijk}, \tau_{ijk}, \delta_{ijk}), \tag{C.1}$$

The distribution is defined by four key parameters: the boundary separation α, which represents the amount of evidence required to make a decision; the bias β, which indicates the initial preference toward one boundary over the other; the non-decision time τ, which accounts for the time unrelated to the decision-making process (e.g., sensory or motor delays); and the drift rate δ, which reflects the quality or speed of evidence accumulation during the task. The parameters of the Wiener distribution could vary across participants and conditions, as well as across trials.

To reduce model complexity, we constrain the model in several ways. First, we treat all parameters as constant across trials. Second, we fixed the β parameter to 0.5α, not allowing it to differ between participants and conditions (i.e., $\beta_{ijk} = \beta$), as we had no theoretical reason to assume a biased starting point. Third, the nondecision time parameter τ was additionally constrained to be constant across conditions (i.e., $\tau_{ijk} = \tau_j$), to simplify the model. We treated differences across individuals as random effects.

To estimate the effect of ERN and CRN on subsequent cognitive adaptation, we regress the drift rate and boundary separation parameters on the EEG amplitude, response type, and condition type as follows,

$$\begin{aligned}\delta_{(ij)} = {} & \theta_{(\delta)j} + \theta_{(c)j} * \mathrm{c}_i + \theta_{(e)j} * \mathrm{e}_i + \theta_{(er)j} * \mathrm{e}_i * \mathrm{r}_i + \\ & + \theta_{(r)} * \mathrm{r}_i + \theta_{(ec)} * \mathrm{e}_i * \mathrm{c}_i + \theta_{(rc)} * \mathrm{r}_i * \mathrm{c}_i + \theta_{(erc)} * \mathrm{e}_i * \mathrm{r}_i * \mathrm{c}_i\end{aligned}$$

$$\begin{aligned}\alpha_{(ij)} = {} & \theta_{(\alpha)j} + \theta_{(c)j} * \mathrm{c}_i + \theta_{(e)j} * \mathrm{e}_i + \theta_{(er)j} * \mathrm{e}_i * \mathrm{r}_i + \\ & + \theta_{(r)} * \mathrm{r}_i + \theta_{(ec)} * \mathrm{e}_i * \mathrm{c}_i + \theta_{(rc)} * \mathrm{r}_i * \mathrm{c}_i + \theta_{(erc)} * \mathrm{e}_i * \mathrm{r}_i * \mathrm{c}_i\end{aligned}$$

where i indicates trial number and j indicates the individual to whom the trial belongs; c_i is a condition type at trial i; r_i is a response type preceding trial i; e_i is a EEG response-related amplitude preceding trial i. θ denotes effects listed in Table 1 on the drift rate (c, e, r stand for condition, EEG amplitude, and response type, respectively).

C.0.1. Priors specification for the drift rate

For random intercept $\theta_{(\delta)j}$ and slopes θ_j, fixed effects, and variability σ, we used the following priors:

$$\begin{aligned}\begin{bmatrix}\theta_{(\delta)j}\\ \theta_{(c)j}\\ \theta_{(e)j}\\ \theta_{(er)j}\end{bmatrix} &\sim \text{Normal}\left(\begin{bmatrix}\mu_{\theta_{(\delta)}}, \sigma_{\theta_{(\delta)}}\\ \mu_{\theta_{(c)}}, \sigma_{\theta_{(c)}}\\ \mu_{\theta_{(e)}}, \sigma_{\theta_{(e)}}\\ \mu_{\theta_{(er)}}, \sigma_{\theta_{(er)}}\end{bmatrix}\right)\\ \mu_{\theta_{(\delta)}}, \mu_{\theta_{(c)}} &\sim \text{Normal}(0, 2)\\ \mu_{\theta_{(e)}}, \mu_{\theta_{(er)}}, \theta_{(r)}, \theta_{(ec)}, \theta_{(rc)}, \theta_{(erc)} &\sim \text{Normal}(0, .5)\\ \sigma_{\theta_{(\delta)}}, \sigma_{\theta_{(c)}} &\sim \Gamma(1, 1)\\ \sigma_{\theta_{(e)}}, \sigma_{\theta_{(er)}} &\sim Normal_{+}(0, .5)\end{aligned} \tag{C.2}$$

C.0.2. Priors specification for the boundary separation

For random intercept $\theta_{(\alpha)j}$ and slopes θ_j, fixed effects, and variability σ, we used the following priors:

$$\begin{aligned}\begin{bmatrix}\theta_{(\alpha)j}\\ \theta_{(c)j}\\ \theta_{(e)j}\\ \theta_{(er)j}\end{bmatrix} &\sim \text{Normal}\left(\begin{bmatrix}\mu_{\theta_{(\alpha)}}, \sigma_{\theta_{(\alpha)}}\\ \mu_{\theta_{(c)}}, \sigma_{\theta_{(c)}}\\ \mu_{\theta_{(e)}}, \sigma_{\theta_{(e)}}\\ \mu_{\theta_{(er)}}, \sigma_{\theta_{(er)}}\end{bmatrix}\right)\\ \mu_{\theta_{(\alpha)}} &\sim \text{Normal}(1, 1)[0, 3]\\ \mu_{\theta_{(c)}} &\sim \text{Normal}(0, 1)\\ \mu_{\theta_{(e)}}, \mu_{\theta_{(er)}}, \theta_{(r)}, \theta_{(ec)}, \theta_{(rc)}, \theta_{(erc)} &\sim \text{Normal}(0, .2)\\ \sigma_{\theta_{(\alpha)}}, \sigma_{\theta_{(c)}} &\sim \Gamma(1, 1)\\ \sigma_{\theta_{(e)}}, \sigma_{\theta_{(er)}} &\sim \text{Normal}_{+}(0, .5)\end{aligned} \tag{C.3}$$

C.0.3. Non-decision time

For the non-decision time we estimated only random intercept. We used the following priors:

$$\begin{aligned}\theta_{(\tau)_j} &\sim \text{Normal}(\mu_{\theta_{(\tau)}}, \sigma_{\theta_{(\tau)}})[0, .4]\\ \mu_{\theta_{(\tau)}} &\sim \text{Normal}(.1, .2)[0, .4]\\ \sigma_{\theta_{(\tau)}} &\sim \Gamma(.3, 1)\end{aligned} \tag{C.4}$$

Appendix D. Model with diffusion coefficient

In the drift diffusion model, the diffusion coefficient quantifies the variability or noise in the evidence accumulation process, which influences both the response time and the accuracy of decisions. A higher diffusion coefficient indicates noisier, less predictable decisions, while a lower diffusion coefficient reflects more consistent and faster decision-making.

Since boundary separation, drift rate, and the diffusion coefficient are not jointly identifiable (see Nunez et al., 2023, for an in-depth discussion), the diffusion coefficient is typically assumed to be constant, with boundary separation and drift rate estimated relative to it. However, if internal noise, as represented by the diffusion coefficient, varies across individuals, this variability could be misattributed to changes in drift rate or boundary separation in models where the diffusion coefficient is held constant. This misattribution could only be revealed by a model that allows for the estimation of the diffusion coefficient parameter. If successful, such a model would provide a more parsimonious interpretation of the results, offering a simpler explanation for the observed data.

D.0.1. Model definition

To test this hypothesis, we aimed to create a drift diffusion model that estimates both fixed and random effects on the diffusion coefficient. To prevent issues with unidentifiability, we fixed boundary separation to 1. The model was defined as follows:

Let Y_{ijk} denote the response vector from the Flanker task, which includes both the response correctness and the response time (X_{ij}, T_{ij}), recorded for the ith trial, $i = 1, \dots, I$, of the kth condition, $k = 1, \dots, K$, of the jth participant, $j = 1, \dots, J$. The observed bivariate data Y_{ijk} is assumed to follow a Wiener distribution, which is described as:

$$Y_{ijk} \sim \text{Wiener}(\alpha_{ijk}/\varsigma_{ijk}, \beta_{ijk}, \tau_{ijk}, \delta_{ijk}/\varsigma_{ijk}), \tag{D.1}$$

The distribution is defined by five key parameters: the boundary separation α, which represents the amount of evidence required to make a decision; the bias β, which indicates the initial preference toward one boundary over the other; the non-decision time τ, which accounts for the time unrelated to the decision-making process (e.g., sensory or motor delays); and the drift rate δ, which reflects the quality or speed of evidence accumulation during the task; and the diffusion coefficient ς, which represents the internal noise in the decision-making process. The parameters of the Wiener distribution could vary across participants and conditions, as well as across trials.

The model was constrained in a several ways: (1) We treat all parameters as constant across trials; (2) The α parameter was fixed to 1, with drift rate δ and diffusion coefficient ς estimated relative to it; (3) the β parameter was fixed to 0.5α, not allowing it to differ between participants and conditions (i.e., $\beta_{ijk} = \beta$), as we had no theoretical reason to assume a biased starting point; (4) the nondecision time parameter τ was additionally constrained to be constant across conditions (i.e., $\tau_{ijk} = \tau_j$), to simplify the model. We treated differences across individuals as random effects.

Additionally, to address issues with fitting ς directly, we reparameterized the model to fit its inversion, i.e., $\kappa = \frac{1}{\varsigma}$. Consequently, our model was defined as follows:

$$Y_{jk} \sim \text{Wiener}(\kappa_{ij}, 0.5, \tau_j, \delta_{ij} * \kappa_{ij}), \tag{D.2}$$

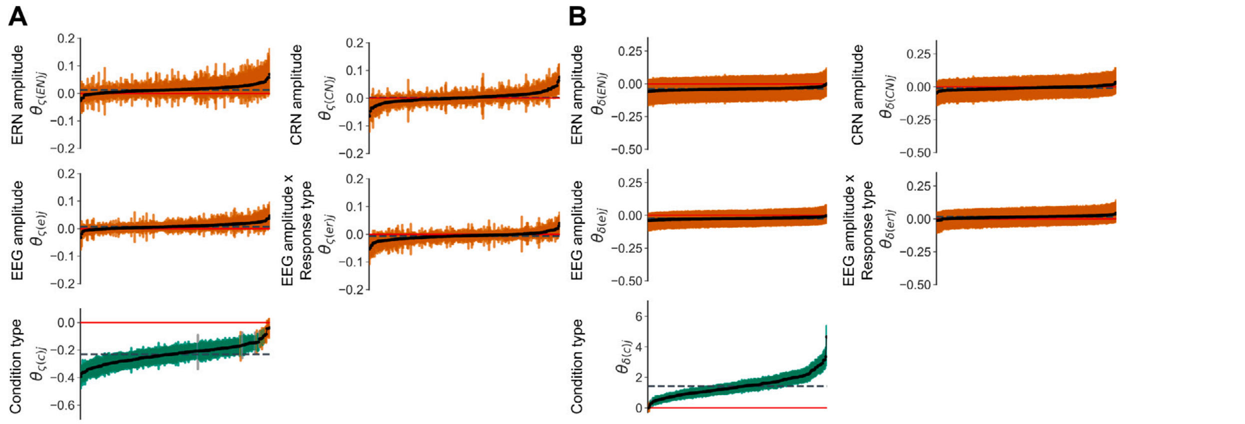

Fig. D.10. Individual EEG amplitudes effects ($\theta_{(e)j}$) and their interactions with response type ($\theta_{(er)j}$), along with the effects of ERN ($\theta_{(EN)j}$), CRN ($\theta_{(CN)j}$), and condition type ($\theta_{(c)j}$) on the diffusion coefficient ς (panel A) and drift rate δ (panel B). The effects are sorted by smallest to largest posterior means for each subplot. The dashed horizontal line denotes the population-level posterior means of the effects. Credible intervals (CrIs) are colored as follows: orange for Bayes Factors less than 0.33, gray for Bayes Factors between 0.33 and 3, and green for Bayes Factors greater than 3.

Table D.8
Posterior means (*b*), standard deviations of the posterior distributions (*SD*), 95% Highest Posterior Density Intervals (*HDIs*), and Bayes Factors (*BF*) of the fixed effects for the diffusion coefficient and the drift rate.

Predictors	Diffusion coefficient ς				Drift rate δ			
	b	*SD*	*HDIs*	*BF*	*b*	*SD*	*HDIs*	*BF*
(Intercept)	0.739	0.012	[0.715, 0.763]	n.a	2.302	0.048	[2.212, 2.399]	n.a
Condition type $\mu_{(c)}$	−0.230	0.011	[−0.251, −0.209]	$>10^{308}$	1.435	0.050	[1.335, 1.533]	$>10^{308}$
EEG amplitude $\mu_{(e)}$	0.008	0.002	[0.004, 0.012]	2.40×10^3	−0.023	0.012	[−0.046, 0.001]	0.145
Response type $\theta_{(r)}$	0.023	0.002	[0.020, 0.027]	$>10^{308}$	−0.039	0.017	[−0.072, −0.005]	0.456
EEG amplitude × Response type $\mu_{(er)}$	−0.006	0.002	[−0.009, −0.002]	0.005	0.015	0.012	[−0.009, 0.039]	0.051
EEG amplitude × Condition type $\theta_{(ec)}$	−0.015	0.004	[−0.022, −0.008]	1.361	0.070	0.012	[0.047, 0.094]	7.69×10^{40}
Response type × Condition type $\theta_{(rc)}$	−0.015	0.004	[−0.022, −0.008]	324	0.262	0.017	[0.229, 0.296]	$>10^{308}$
ERN amplitude $\mu_{(EN)}$	0.013	0.003	[0.007, 0.020]	1.56	−0.038	0.021	[−0.080, 0.004]	0.209
CRN amplitude $\mu_{(CN)}$	0.002	0.002	[−0.001, 0.006]	8.72×10^{-5}	−0.009	0.012	[−0.032, 0.014]	0.030

Note. c, *e*, and *r* denote condition, ERN/CRN amplitude, and response type, respectively. Detailed descriptions of the predictors are provided in Table 1. Bayes Factor was calculated as Savage–Dickey ratio. *BF* above 5 indicates an evidence for an effect, *BF* between 5 and . 20 suggests insufficient evidence to determine the presence of the effect, and *BF* lower than . 20 indicates strong evidence for the null effect.

To estimate the effect of ERN and CRN on cognitive processes, we regress the inverted diffusion coefficient κ and drift rate δ parameters onto the EEG amplitude, response type, and condition type as follows,

$$\delta_{(ij)} = \theta_{(\delta)j} + \theta_{(c)j} * c_i + \theta_{(e)j} * e_i + \theta_{(er)j} * e_i * r_i + \\ + \theta_{(r)} * r_i + \theta_{(ec)} * e_i * c_i + \theta_{(rc)} * r_i * c_i + \theta_{(erc)} * e_i * r_i * c_i$$

$$\kappa_{(ij)} = \theta_{(\kappa)_j} + \theta_{(c)j} * c_i + \theta_{(e)j} * e_i + \theta_{(er)j} * e_i * r_i + \\ + \theta_{(r)} * r_i + \theta_{(ec)} * e_i * c_i + \theta_{(rc)} * r_i * c_i + \theta_{(erc)} * e_i * r_i * c_i$$

where i indicates trial number and j indicates the individual to whom the trial belongs; c_i is a condition type at trial i; r_i is a response type preceding trial i; e_i is a EEG response-related amplitude preceding trial i. θ denotes effects listed in Table 1 and *c*, *e*, and *r* stand for condition, EEG amplitude, and response type, respectively.

The priors for all effects on κ were identical to those specified for boundary separation in the main analysis (see Appendix C). Similarly, the priors for δ, and τ were consistent with those used in the main analysis.

The effects of EEG amplitude, response type, and condition type on the diffusion coefficient ς were estimated using appropriate transformations of the effects on κ (see the model code at https://github.com/abelowska/jointErrorCMD).

D.0.2. Results

The diffusion coefficient model demonstrated satisfactory convergence and efficiency statistics. Most parameters achieved $\hat{R}$ values below 1.01 and effective sample sizes exceeding 10,000. The highest $\hat{R}$ was 1.04, and the lowest effective sample size was 160.22, observed for the between-subject variability parameter of the EEG amplitude and response type interaction effect $\theta_{\delta er(j)}$ effect. Detailed results on model convergence are available on OSF.

At the population level, the effect of EEG amplitude was significantly associated only with the diffusion coefficient; however, this effect was very small (b = .008, 95% HDI [0.004, 0.012], BF = 2.40×10^3). Thus, more pronounced EEG amplitudes were associated with slightly less noisy evidence accumulation in the subsequent trial. Both the effect of EEG amplitude on the drift rate and the interaction effect of EEG amplitude and response type were non-significant, with Bayes Factors providing evidence for the null. A summary of the estimated fixed effects is presented in Table D.8.

Figs. D.10 visualizes the between-participants variability in the effects of ERN and CRN amplitudes on diffusion coefficient and drift rate. Specifically, it shows the posterior means of the individual slopes for EEG signals ($\theta_{\varsigma(e)j}$ and $\theta_{\delta(e)j}$), interaction effect of EEG and response type ($\theta_{\varsigma(er)j}$ and $\theta_{\delta(er)j}$), ERN amplitudes ($\theta_{\varsigma(EN)j}$ and $\theta_{\delta(EN)j}$) and CRN amplitudes ($\theta_{\varsigma(CN)j}$ and $\theta_{\delta(CN)j}$), along with their corresponding credible intervals (CrIs).

There was significant variability in the effects of EEG amplitude and the interaction between EEG amplitude and response type, and thus the effects of ERN and CRN amplitudes, on the diffusion coefficient across participants. Some individuals showed positive effects, while others displayed negative effects. However, Bayes Factors and posterior distributions indicated that these random effects were non-significant, providing evidence for null effects.

The effects of ERN and CRN amplitudes on the drift rate followed the trend observed in the main analysis. However, the reduced strength of these effects (estimated relative to the boundary), yielded them non-significant, with Bayes Factors supporting the null hypothesis.

Overall, given the very small effect size of ERN and CRN amplitudes on the diffusion coefficient, we cannot attribute the relationship between ERN and CRN amplitudes and DDM parameters in the subsequent trial to changes in the participant's internal noise. Nonetheless, since this relationship, however small, is evident at the group level, we suggest that the diffusion coefficient should always be considered in such analyses.

Data availability

The code to reproduce all analyses is available at https://github.com/abelowska/jointErrorCMD The dataset, consisting of EEG, behavioral, and questionnaire data is available at OSF https://osf.io/jszyh/.

References

Allain, S., Burle, B., Hasbroucq, T., Vidal, F., 2009. Sequential adjustments before and after partial errors. Psychon. Bull. Rev. 16 (2), 356–362. http://dx.doi.org/10.3758/PBR.16.2.356.

Bartholow, B.D., Pearson, M.A., Dickter, C.L., Sher, K.J., Fabiani, M., Gratton, G., 2005. Strategic control and medial frontal negativity: beyond errors and response conflict. Psychophysiology 42 (1), 33–42. http://dx.doi.org/10.1111/j.1469-8986.2005.00258.x.

Beatty, P.J., Buzzell, G.A., Roberts, D.M., Voloshyna, Y., McDonald, C.G., 2021. Subthreshold error corrections predict adaptive post-error compensations. Psychophysiology 58 (6), e13803. http://dx.doi.org/10.1111/psyp.13803.

Betancourt, M., 2016. Identifying the Optimal Integration Time in Hamiltonian Monte Carlo. http://dx.doi.org/10.48550/arXiv.1601.00225.

Betancourt, M., 2018. A conceptual introduction to Hamiltonian Monte Carlo. http://dx.doi.org/10.48550/arXiv.1701.02434.

BMJ, 1996. Declaration of Helsinki (1964). BMJ 313 (7070), 1448–1449. http://dx.doi.org/10.1136/bmj.313.7070.1448a.

Boehm, U., Annis, J., Frank, M.J., Hawkins, G.E., Heathcote, A., Kellen, D., Krypotos, A.-M., Lerche, V., Logan, G.D., Palmeri, T.J., van Ravenzwaaij, D., Servant, M., Singmann, H., Starns, J.J., Voss, A., Wiecki, T.V., Matzke, D., Wagenmakers, E.J., 2018. Estimating across-trial variability parameters of the Diffusion Decision Model: Expert advice and recommendations. J. Math. Psych. 87, 46–75. http://dx.doi.org/10.1016/j.jmp.2018.09.004.

Boksem, M.A.S., Meijman, T.F., Lorist, M.M., 2006. Mental fatigue, motivation and action monitoring. Biol. Psychol. 72 (2), 123–132. http://dx.doi.org/10.1016/j.biopsycho.2005.08.007.

Bonini, F., Burle, B., Liégeois-Chauvel, C., Régis, J., Chauvel, P., Vidal, F., 2014. Action monitoring and medial frontal cortex: leading role of supplementary motor area. Sci. (N. Y. N. Y.) 343 (6173), 888–891. http://dx.doi.org/10.1126/science.1247412.

Botvinick, M.M., Braver, T.S., Barch, D.M., Carter, C.S., Cohen, J.D., 2001. Conflict monitoring and cognitive control. Psychol Rev 108 (3), 624–652. http://dx.doi.org/10.1037/0033-295X.108.3.624, Place: US Publisher: American Psychological Association.

Carleton, R.N., Norton, M.A.P.J., Asmundson, G.J.G., 2007. Fearing the unknown: A short version of the Intolerance of Uncertainty Scale. J. Anxiety Disord. 21 (1), 105–117. http://dx.doi.org/10.1016/j.janxdis.2006.03.014.

Carver, C.S., White, T.L., 1994. Behavioral inhibition, behavioral activation, and affective responses to impending reward and punishment: The BIS/BAS Scales. J. Pers. Soc. Psychol. 67, 319–333. http://dx.doi.org/10.1037/0022-3514.67.2.319.

Cavanagh, J.F., Frank, M.J., 2014. Frontal theta as a mechanism for cognitive control. Trends Cogn. Sci. 18 (8), 414–421. http://dx.doi.org/10.1016/j.tics.2014.04.012.

Cavanagh, J.F., Shackman, A.J., 2015. Frontal midline theta reflects anxiety and cognitive control: meta-analytic evidence. J. Physiol. Paris 109 (1–3), 3–15. http://dx.doi.org/10.1016/j.jphysparis.2014.04.003.

Clayson, P.E., Brush, C.J., Hajcak, G., 2021. Data quality and reliability metrics for event-related potentials (ERPs): The utility of subject-level reliability. Int. J. Psychophysiol. 165, 121–136. http://dx.doi.org/10.1016/j.ijpsycho.2021.04.004.

Clayson, P.E., Miller, G.A., 2017. ERP Reliability Analysis (ERA) Toolbox: An open-source toolbox for analyzing the reliability of event-related brain potentials. Int. J. Psychophysiol. 111, 68–79. http://dx.doi.org/10.1016/j.ijpsycho.2016.10.012.

Crump, M.J.C., Logan, G.D., 2013. Prevention and correction in post-error performance: An ounce of prevention, a pound of cure. J. Exp. Psychol. [Gen.] 142 (3), 692–709. http://dx.doi.org/10.1037/a0030014, Place: US Publisher: American Psychological Association.

Danielmeier, C., Eichele, T., Forstmann, B.U., Tittgemeyer, M., Ullsperger, M., 2011. Posterior medial frontal cortex activity predicts post-error adaptations in task-related visual and motor areas. J. Neurosci. 31 (5), 1780–1789. http://dx.doi.org/10.1523/JNEUROSCI.4299-10.2011.

Danielmeier, C., Ullsperger, M., 2011. Post-error adjustments. Front. Psychol. 2, 233. http://dx.doi.org/10.3389/fpsyg.2011.00233.

Danielmeier, C., Wessel, J.R., Steinhauser, M., Ullsperger, M., 2009. Modulation of the error-related negativity by response conflict. Psychophysiology 46 (6), 1288–1298. http://dx.doi.org/10.1111/j.1469-8986.2009.00860.x, Place: United Kingdom Publisher: Wiley-Blackwell Publishing Ltd..

Debener, S., Ullsperger, M., Siegel, M., Fiehler, K., von Cramon, D.Y., Engel, A.K., 2005. Trial-by-trial coupling of concurrent electroencephalogram and functional magnetic resonance imaging identifies the dynamics of performance monitoring. J. Neurosci.: Off. J. Soc. Neurosci. 25 (50), 11730–11737. http://dx.doi.org/10.1523/JNEUROSCI.3286-05.2005.

Dickey, J.M., 1971. The weighted likelihood ratio, linear hypotheses on normal location parameters. Ann. Math. Stat. 42 (1), 204–223.

Dutilh, G., Vandekerckhove, J., Forstmann, B.U., Keuleers, E., Brysbaert, M., Wagenmakers, E.-J., 2012. Testing theories of post-error slowing. Atten. Percept., & Psychophys. 74 (2), 454–465. http://dx.doi.org/10.3758/s13414-011-0243-2.

Endrass, T., Klawohn, J., Gruetzmann, R., Ischebeck, M., Kathmann, N., 2012. Response-related negativities following correct and incorrect responses: Evidence from a temporospatial principal component analysis. Psychophysiology 49 (6), 733–743. http://dx.doi.org/10.1111/j.1469-8986.2012.01365.x.

Endrass, T., Klawohn, J., Schuster, F., Kathmann, N., 2008. Overactive performance monitoring in obsessive-compulsive disorder: ERP evidence from correct and erroneous reactions. Neuropsychologia 46 (7), 1877–1887. http://dx.doi.org/10.1016/j.neuropsychologia.2007.12.001.

Eriksen, B.A., Eriksen, C.W., 1974. Effects of noise letters upon the identification of a target letter in a nonsearch task. Percept. Psychophys. 16 (1), 143–149. http://dx.doi.org/10.3758/BF03203267, Place: US Publisher: Psychonomic Society.

Falkenstein, M., Hohnsbein, J., Hoormann, J., Blanke, L., 1991. Effects of crossmodal divided attention on late ERP components. II. Error processing in choice reaction tasks. Electroencephalogr. Clin. Neurophysiol. 78 (6), 447–455. http://dx.doi.org/10.1016/0013-4694(91)90062-9.

Ficarella, S.C., Rochet, N., Burle, B., 2019. Becoming aware of subliminal responses: An EEG/EMG study on partial error detection and correction in humans. Cortex 120, 443–456. http://dx.doi.org/10.1016/j.cortex.2019.07.007.

Fiehler, K., Ullsperger, M., Von Cramon, D.Y., 2005. Electrophysiological correlates of error correction. Psychophysiology 42 (1), 72–82. http://dx.doi.org/10.1111/j.1469-8986.2005.00265.x, Place: United Kingdom Publisher: Blackwell Publishing.

Files, B.T., Pollard, K.A., Oiknine, A.H., Khooshabeh, P., Passaro, A.D., 2021. Correct response negativity may reflect subjective value of reaction time under regulatory fit in a speed-rewarded task. Psychophysiology 58 (9), e13856. http://dx.doi.org/10.1111/psyp.13856.

Fischer, A.G., Danielmeier, C., Villringer, A., Klein, T.A., Ullsperger, M., 2016. Gender influences on brain responses to errors and post-error adjustments. Sci. Rep. 6 (1), 24435. http://dx.doi.org/10.1038/srep24435.

Fischer, A.G., Klein, T.A., Ullsperger, M., 2017. Comparing the error-related negativity across groups: The impact of error- and trial-number differences. Psychophysiology 54 (7), 998–1009. http://dx.doi.org/10.1111/psyp.12863.

Fisher, A.J., Medaglia, J.D., Jeronimus, B.F., 2018. Lack of group-to-individual generalizability is a threat to human subjects research. Proc. Natl. Acad. Sci. USA 115 (27), E6106–E6115. http://dx.doi.org/10.1073/pnas.1711978115.

Foa, E.B., Huppert, J.D., Leiberg, S., Langner, R., Kichic, R., Hajcak, G., Salkovskis, P.M., 2002. The Obsessive-Compulsive Inventory: development and validation of a short version. Psychol. Assess. 14 (4), 485–496.

Frank, M.J., Woroch, B.S., Curran, T., 2005. Error-related negativity predicts reinforcement learning and conflict biases. Neuron 47 (4), 495–501. http://dx.doi.org/10.1016/j.neuron.2005.06.020.

Frömer, R., Lin, H., Dean Wolf, C.K., Inzlicht, M., Shenhav, A., 2021. Expectations of reward and efficacy guide cognitive control allocation. Nat. Commun. 12 (1), 1030. http://dx.doi.org/10.1038/s41467-021-21315-z.

Frömer, R., Shenhav, A., 2022. Filling the gaps: Cognitive control as a critical lens for understanding mechanisms of value-based decision-making. Neurosci. Biobehav. Rev. 134, 104483. http://dx.doi.org/10.1016/j.neubiorev.2021.12.006.

Fu, Z., Wu, D.-A.J., Ross, I., Chung, J.M., Mamelak, A.N., Adolphs, R., Rutishauser, U., 2019. Single-neuron correlates of error monitoring and post-error adjustments in human medial frontal cortex. Neuron 101 (1), 165–177.e5. http://dx.doi.org/10.1016/j.neuron.2018.11.016.

Gehring, W.J., Goss, B., Coles, M.G.H., Meyer, D.E., Donchin, E., 1993. A neural system for error detection and compensation. Psychol. Sci. 4 (6), 385–390. http://dx.doi.org/10.1111/j.1467-9280.1993.tb00586.x.

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B., 2015. Bayesian Data Analysis, third ed. Chapman and Hall/CRC, New York, http://dx.doi.org/10.1201/b16018.

Grabowska, A., Sondej, F., Senderecka, M., 2024a. A network analysis of affective and motivational individual differences and error monitoring in a non-clinical sample. Cerebral Cortex 34 (10), bhae397. http://dx.doi.org/10.1093/cercor/bhae397.

Grabowska, A., Zabielski, J., Senderecka, M., 2024b. Machine learning reveals differential effects of depression and anxiety on reward and punishment processing. Sci. Rep. 14 (1), 8422. http://dx.doi.org/10.1038/s41598-024-58031-9.

Gramfort, A., Luessi, M., Larson, E., Engemann, D.A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., Hämäläinen, M., 2013. MEG and EEG data analysis with MNE-Python. Front. Neurosci. 7, 1–13. http://dx.doi.org/10.3389/fnins.2013.00267.

Gratton, G., Coles, M.G.H., Donchin, E., 1983. A new method for off-line removal of ocular artifact. Electroencephalogr. Clin. Neurophysiol. 55 (4), 468–484. http://dx.doi.org/10.1016/0013-4694(83)90135-9.

Gratton, G., Coles, M.G., Donchin, E., 1992. Optimizing the use of information: strategic control of activation of responses. J. Exp. Psychol. Gen. 121 (4), 480–506. http://dx.doi.org/10.1037//0096-3445.121.4.480.

Grupe, D.W., Nitschke, J.B., 2013. Uncertainty and anticipation in anxiety: an integrated neurobiological and psychological perspective. Nature Rev. Neurosci. 14 (7), 488–501. http://dx.doi.org/10.1038/nrn3524.

Haaf, J.M., Rouder, J.N., 2017. Developing constraint in bayesian mixed models. Psychol. Methods 22 (4), 779–798. http://dx.doi.org/10.1037/met0000156.

Haaf, J.M., Rouder, J.N., 2018. Some do and some don't? Accounting for variability of individual difference structures. Psychon. Bull. Rev. 26 (3), 772–789. http://dx.doi.org/10.3758/s13423-018-1522-x.

Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M.H., Brett, M., Haldane, A., del Río, J.F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., Oliphant, T.E., 2020. Array programming with NumPy. Nature 585 (7825), 357–362. http://dx.doi.org/10.1038/s41586-020-2649-2.

Hedge, C., Vivian-Griffiths, S., Powell, G., Bompas, A., Sumner, P., 2019. Slow and steady? Strategic adjustments in response caution are moderately reliable and correlate across tasks. Conscious. Cogn. 75, 102797. http://dx.doi.org/10.1016/j.concog.2019.102797.

Hoffman, M., Gelman, A., 2011. The No-U-turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. J. Mach. Learn. Res..

Holroyd, C.B., Coles, M.G.H., 2002. The neural basis of human error processing: Reinforcement learning, dopamine, and the error-related negativity. Psychol Rev 109, 679–709. http://dx.doi.org/10.1037/0033-295X.109.4.679.

Hu, L., Mouraux, A., Hu, Y., Iannetti, G.D., 2010. A novel approach for enhancing the signal-to-noise ratio and detecting automatically event-related potentials (ERPs) in single trials. NeuroImage 50 (1), 99–111. http://dx.doi.org/10.1016/j.neuroimage.2009.12.010.

Huppert, J.D., Walther, M.R., Hajcak, G., Yadin, E., Foa, E.B., Simpson, H.B., Liebowitz, M.R., 2007. The OCI-R: Validation of the subscales in a clinical sample. J. Anxiety Disord. 21 (3), 394–406. http://dx.doi.org/10.1016/j.janxdis.2006.05.006.

Imhof, M.F., Rüsseler, J., 2019. Performance monitoring and correct response significance in conscientious individuals. Front. Hum. Neurosci. 13, http://dx.doi.org/10.3389/fnhum.2019.00239.

Janowski, K., 2007. Kwestionariusz Oceny Martwienia sie, polska adaptacja PSWQ [The PSWQ worry questionnaire. Polish adaptation]. Wydawnictwo KUL.

Jentzsch, I., Dudschig, C., 2009. Short Article: Why do we slow down after an error? Mechanisms underlying the effects of posterror slowing. Q. J. Exp. Psychol. 62 (2), 209–218. http://dx.doi.org/10.1080/17470210802240655, Publisher: SAGE Publications.

Kalfaoglu, C., Stafford, T., Milne, E., 2018. Frontal theta band oscillations predict error correction and posterror slowing in typing. J. Exp. Psychol. [Hum. Percept.] 44 (1), 69–88. http://dx.doi.org/10.1037/xhp0000417.

Kopp, B., Rist, F., Mattler, U., 1996. N200 in the flanker task as a neurobehavioral tool for investigating executive control. Psychophysiology 33 (3), 282–294. http://dx.doi.org/10.1111/j.1469-8986.1996.tb00425.x.

Kossowska, M., 2003. Roznice indywidualne w potrzebie poznawczego domkniecia. Przeglad Psychol. 46 (4).

Kumar, R., Carroll, C., Hartikainen, A., Martin, O., 2019. Arviz a unified library for exploratory analysis of Bayesian models in python. J. Open Source Softw. 4 (33), 1143. http://dx.doi.org/10.21105/joss.01143.

Laming, D., 1968. Information Theory of Choice-Reaction Times. Academic Press, Oxford, England.

Laming, D., 1979. Choice reaction performance following an error. Acta Psychol. 43 (3), 199–224. http://dx.doi.org/10.1016/0001-6918(79)90026-X.

Lee, M.D., Newell, B.J., 2011. Using hierarchical Bayesian methods to examine the tools of decision-making. Judgm. Decis. Mak. 6 (8), 832–842. http://dx.doi.org/10.1017/S1930297500004253, Place: US Publisher: Society for Judgment and Decision Making.

Lerche, V., Voss, A., 2016. Model complexity in diffusion modeling: Benefits of making the model more parsimonious. Front. Psychol. 7, http://dx.doi.org/10.3389/fpsyg.2016.01324.

Lilly, J.M., 2017. Element analysis: a wavelet-based method for analysing time-localized events in noisy time series. Proc. R. Soc. A: Math. Phys. Eng. Sci. 473 (2200), 20160776. http://dx.doi.org/10.1098/rspa.2016.0776.

Lüken, M., Heathcote, A., Haaf, J.M., Matzke, D., 2023. Parameter Identifiability in Evidence-Accumulation Models: The Effect of Error Rates on the Diffusion Decision Model and the Linear Ballistic Accumulator. OSF, http://dx.doi.org/10.31234/osf.io/wsgnt.

Luu, P., Collins, P., Tucker, D.M., 2000. Mood, personality, and self-monitoring: Negative affect and emotionality in relation to frontal lobe mechanisms of error monitoring. J. Exp. Psychol. [Gen.] 129 (1), 43–60. http://dx.doi.org/10.1037/0096-3445.129.1.43, Place: US Publisher: American Psychological Association.

Marco-Pallarés, J., Camara, E., Münte, T.F., Rodríguez-Fornells, A., 2008. Neural mechanisms underlying adaptive actions after slips. J. Cogn. Neurosci. 20 (9), 1595–1610. http://dx.doi.org/10.1162/jocn.2008.20117.

Maruo, Y., Masaki, H., 2022. A possibility of error-related processing contamination in the No-go N2: The effect of partial-error trials on response inhibition processing. Eur. J. Neurosci. 55 (8), 1934–1946. http://dx.doi.org/10.1111/ejn.15658.

Maruo, Y., Sommer, W., Masaki, H., 2017. The effect of monetary punishment on error evaluation in a Go/No-go task. Int. J. Psychophysiol. 120, 54–59. http://dx.doi.org/10.1016/j.ijpsycho.2017.07.002, Place: Netherlands Publisher: Elsevier Science.

Matsuhashi, T., Segalowitz, S.J., Murphy, T.I., Nagano, Y., Hirao, T., Masaki, H., 2021. Medial frontal negativities predict performance improvements during motor sequence but not motor adaptation learning. Psychophysiology 58 (1), e13708. http://dx.doi.org/10.1111/psyp.13708.

Mattes, A., Porth, E., Niessen, E., Kummer, K., Mück, M., Stahl, J., 2023. Larger error negativity peak amplitudes for accuracy versus speed instructions may reflect more neuro-cognitive alignment, not more intense error processing. Sci. Rep. 13 (1), 2259. http://dx.doi.org/10.1038/s41598-023-29434-x.

Mattes, A., Porth, E., Stahl, J., 2022. Linking neurophysiological processes of action monitoring to post-response speed-accuracy adjustments in a neuro-cognitive diffusion model. NeuroImage 247, 118798. http://dx.doi.org/10.1016/j.neuroimage.2021.118798.

Meyer, T.J., Miller, M.L., Metzger, R.L., Borkovec, T.D., 1990. Development and validation of the penn state worry questionnaire. Behav. Res. Ther. 28 (6), 487–495. http://dx.doi.org/10.1016/0005-7967(90)90135-6.

Müller, J.M., Wytykowska, A.M., 2005. Psychometric properties and validation of a Polish adaptation of Carver and White's BIS/BAS scales. Pers. Individ. Differ. 4 (39), 795–805. http://dx.doi.org/10.1016/j.paid.2005.03.006.

van Noordt, S., Desjardins, J., Segalowitz, S., 2015. Watch out! medial frontal cortex is activated by cues signaling potential changes in response demands. NeuroImage 114, http://dx.doi.org/10.1016/j.neuroimage.2015.04.021.

Notebaert, W., Houtman, F., Opstal, F.V., Gevers, W., Fias, W., Verguts, T., 2009. Post-error slowing: An orienting account. Cognition 111 (2), 275–279. http://dx.doi.org/10.1016/j.cognition.2009.02.002.

Nunez, M.D., Fernandez, K., Srinivasan, R., Vandekerckhove, J., 2024. A tutorial on fitting joint models of M/EEG and behavior to understand cognition. Behav. Res. Methods 56 (6), 6020–6050. http://dx.doi.org/10.3758/s13428-023-02331-x.

Nunez, P.L., Nunez, M.D., Srinivasan, R., 2019. Multi-scale neural sources of EEG: Genuine, equivalent, and representative. A tutorial review. Brain Topogr. 32 (2), 193–214. http://dx.doi.org/10.1007/s10548-019-00701-3.

Nunez, M.D., Schubert, A.L., Frischkorn, G.T., Oberauer, K., 2023. Cognitive Models of Decision-Making with Identifiable Parameters: Diffusion Decision Models with Within-Trial Noise. OSF, http://dx.doi.org/10.31234/osf.io/h4fde.

Nunez, M.D., Vandekerckhove, J., Srinivasan, R., 2017. How attention influences perceptual decision making: Single-trial EEG correlates of drift-diffusion model parameters. J. Math. Psych. 76 (Pt B), 117–130. http://dx.doi.org/10.1016/j.jmp.2016.03.003.

Pailing, P., Segalowitz, S., 2004. The effects of uncertainty in error monitoring on associated ERPs. Brain Cogn. 56, 215–233. http://dx.doi.org/10.1016/j.bandc.2004.06.005.

Palestro, J.J., Bahg, G., Sederberg, P.B., Lu, Z.L., Steyvers, M., Turner, B.M., 2018. A tutorial on joint models of neural and behavioral measures of cognition. J. Math. Psych. 84, 20–48. http://dx.doi.org/10.1016/j.jmp.2018.03.003.

Park, B., Holbrook, A., Lutz, M.C., Baldwin, S.A., Larson, M.J., Clayson, P.E., 2024. Task-specific relationships between error-related ERPs and behavior: Flanker, Stroop, and Go/Nogo tasks. Int. J. Psychophysiol.: Off. J. Int. Organ. Psychophysiol. 204, 112409. http://dx.doi.org/10.1016/j.ijpsycho.2024.112409.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011. Scikit-learn: Machine learning in Python. J. Mach. Learn. Res. 12 (85), 2825–2830.

Peirce, J.W., 2007. PsychoPy—Psychophysics software in Python. J. Neurosci. Methods 162 (1–2), 8–13. http://dx.doi.org/10.1016/j.jneumeth.2006.11.017.

Quiroga, R.Q., Garcia, H., 2003. Single-trial event-related potentials with wavelet denoising. Clin. Neurophysiol. 114 (2), 376–390. http://dx.doi.org/10.1016/S1388-2457(02)00365-6.

Rabbitt, P.M., 1966. Errors and error correction in choice-response tasks. J. Exp. Psychol. 71 (2), 264–272. http://dx.doi.org/10.1037/h0022853, Place: US Publisher: American Psychological Association.

Rabbitt, P., 1979. How old and young subjects monitor and control responses for accuracy and speed. Br. J. Psychol. 70 (2), 305–311. http://dx.doi.org/10.1111/j.2044-8295.1979.tb01687.x.

Rabbitt, P., Rodgers, B., 1977. What does a man do after he makes an error? an analysis of response programming. Q. J. Exp. Psychol. http://dx.doi.org/10.1080/14640747708400645.

Ratcliff, R., McKoon, G., 2008. The diffusion decision model: Theory and data for two-choice decision tasks. Neural Comput. 20 (4), 873–922. http://dx.doi.org/10.1162/neco.2008.12-06-420.

Riesel, A., Endrass, T., Kaufmann, C., Kathmann, N., 2011. Overactive error-related brain activity as a candidate endophenotype for obsessive-compulsive disorder: evidence from unaffected first-degree relatives. Am. J. Psychiatry 168 (3), 317–324. http://dx.doi.org/10.1176/appi.ajp.2010.10030416.

Riesel, A., Härpfer, K., Thoma, L., Kathmann, N., Klawohn, J., 2023. Associations of neural error-processing with symptoms and traits in a dimensional sample recruited across the obsessive-compulsive spectrum. Psychophysiology 60 (2), e14164. http://dx.doi.org/10.1111/psyp.14164.

Roger, C., Bénar, C.G., Vidal, F., Hasbroucq, T., Burle, B., 2010. Rostral Cingulate Zone and correct response monitoring: ICA and source localization evidences for the unicity of correct- and error-negativities. NeuroImage 51 (1), 391–403. http://dx.doi.org/10.1016/j.neuroimage.2010.02.005.

Rouder, J.N., Haaf, J.M., 2021. Are there reliable qualitative individual difference in cognition? J. Cogn. 4 (1), 46. http://dx.doi.org/10.5334/joc.131.

Rouder, J.N., Lu, J., 2005. An introduction to Bayesian hierarchical models with an application in the theory of signal detection. Psychon. Bull. Rev. 12 (4), 573–604. http://dx.doi.org/10.3758/BF03196750, Place: US Publisher: Psychonomic Society.

Scheffers, M.K., Coles, M.G.H., 2000. Performance monitoring in a confusing world: error-related brain activity, judgments of response accuracy, and types of errors. J. Exp. Psychol. Hum. Percept. Perform. 26 (1), 141–151. http://dx.doi.org/10.1037//0096-1523.26.1.141.

Schiffler, B.C., Bengtsson, S.L., Lundqvist, D., 2017. The sustained influence of an error on future decision-making. Front. Psychol. 8, http://dx.doi.org/10.3389/fpsyg.2017.01077.

Senderecka, M., Szewczyk, J., Wichary, S., Kossowska, M., 2018. Individual differences in decisiveness: ERP correlates of response inhibition and error monitoring. Psychophysiology 55 (10), e13198. http://dx.doi.org/10.1111/psyp.13198.

Shackman, A.J., Salomons, T.V., Slagter, H.A., Fox, A.S., Winter, J.J., Davidson, R.J., 2011. The integration of negative affect, pain and cognitive control in the cingulate cortex. Nature Rev. Neurosci. 12 (3), 154–167. http://dx.doi.org/10.1038/nrn2994.

Spielberger, C.D., Gorsuch, R.L., Lushene, R.E., 1970. STAI manual for the Stait-Trait Anxiety Inventory (“self-evaluation questionnaire”). Consulting Psychologists Press, Palo Alto, Calif.

Spielberger, C.D., Strelau, J., Wrześniewski, K., Tysarczyk, M., 2006. STAI - Inwentarz Stanu i Cechy Leku. Pracownia Testów Psychologicznych, Warszawa.

Stan Development Team, 2021. CmdStanPy, 0.9.76.

Stan Development Team, 2024. Stan Modeling Language Users Guide and Reference Manual, 2.35.

Thakur, G., Brevdo, E., Fučkar, N.S., Wu, H.T., 2013. The Synchrosqueezing algorithm for time-varying spectral analysis: Robustness properties and new paleoclimate applications. Signal Process. 93 (5), 1079–1094. http://dx.doi.org/10.1016/j.sigpro.2012.11.029.

Torrence, C., Compo, G.P., 1998. A practical guide to wavelet analysis. Bull. Am. Meteorol. Soc. 79 (1), 61–78. http://dx.doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2.

Ullsperger, M., 2024. Beyond peaks and troughs: Multiplexed performance monitoring signals in the EEG. Psychophysiology 61 (7), e14553. http://dx.doi.org/10.1111/psyp.14553.

Ullsperger, M., von Cramon, D.Y., 2001. Subprocesses of performance monitoring: A dissociation of error processing and response competition revealed by event-related fMRI and ERPs. NeuroImage 14 (6), 1387–1401. http://dx.doi.org/10.1006/nimg.2001.0935.

Ullsperger, M., Danielmeier, C., Jocham, G., 2014. Neurophysiology of performance monitoring and adaptive behavior. Physiol. Rev. 94 (1), 35–79. http://dx.doi.org/10.1152/physrev.00041.2012.

Van Noordt, S.J., Campopiano, A., Segalowitz, S.J., 2016. A functional classification of medial frontal negativity ERPs: Theta oscillations and single subject effects. Psychophysiology 53 (9), 1317–1334. http://dx.doi.org/10.1111/psyp.12689.

Vandekerckhove, J., Tuerlinckx, F., Lee, M.D., 2011. Hierarchical diffusion models for two-choice response times. Psychol. Methods 16 (1), 44–62. http://dx.doi.org/10.1037/a0021765.

Vidal, F., Burle, B., Bonnet, M., Grapperon, J., Hasbroucq, T., 2003. Error negativity on correct trials: a reexamination of available data. Biol. Psychol. 64 (3), 265–282. http://dx.doi.org/10.1016/S0301-0511(03)00097-8.

Vidal, F., Burle, B., Hasbroucq, T., 2022. On the comparison between the Nc/CRN and the Ne/ERN. Front. Hum. Neurosci. 15, 788167. http://dx.doi.org/10.3389/fnhum.2021.788167.

Vidal, F., Hasbroucq, T., Grapperon, J., Bonnet, M., 2000. Is the ‘error negativity’ specific to errors? Biol. Psychol. 51 (2), 109–128. http://dx.doi.org/10.1016/S0301-0511(99)00032-0.

Voss, A., Nagler, M., Lerche, V., 2013. Diffusion models in experimental psychology. Exp. Psychol. 60 (6), 385–402. http://dx.doi.org/10.1027/1618-3169/a000218.

Voss, A., Rothermund, K., Voss, J., 2004. Interpreting the parameters of the diffusion model: An empirical validation. Mem. Cogn. 32 (7), 1206–1220. http://dx.doi.org/10.3758/BF03196893.

Voss, A., Voss, J., Lerche, V., 2015. Assessing cognitive processes with diffusion model analyses: a tutorial based on fast-dm-30. Front. Psychol. 6, 336. http://dx.doi.org/10.3389/fpsyg.2015.00336.

Wagenmakers, E.J., Lodewyckx, T., Kuriyal, H., Grasman, R., 2010. Bayesian hypothesis testing for psychologists: a tutorial on the Savage-Dickey method. Cogn. Psychol. 60 (3), 158–189. http://dx.doi.org/10.1016/j.cogpsych.2009.12.001.

Watanabe, S., 2010. Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory. J. Mach. Learn. Res. 11 (116), 3571–3594.

Webster, D.M., Kruglanski, A.W., 1994. Individual differences in need for cognitive closure. J. Pers. Soc. Psychol. 67 (6), 1049–1062. http://dx.doi.org/10.1037/0022-3514.67.6.1049, Place: US Publisher: American Psychological Association.

Weinberg, A., Dieterich, R., Riesel, A., 2015. Error-related brain activity in the age of RDoC: A review of the literature. Int. J. Psychophysiol. 98 (2, Part 2), 276–299. http://dx.doi.org/10.1016/j.ijpsycho.2015.02.029.

Weinberg, A., Meyer, A., Hale-Rude, E., Perlman, G., Kotov, R., Klein, D.N., Hajcak, G., 2016. Error-related negativity (ERN) and sustained threat: Conceptual framework and empirical evaluation in an adolescent sample. Psychophysiology 53 (3), 372–385. http://dx.doi.org/10.1111/psyp.12538.

Wessel, J.R., 2018. An adaptive orienting theory of error processing. Psychophysiology 55 (3), e13041. http://dx.doi.org/10.1111/psyp.13041.

Wessel, J.R., Aron, A.R., 2017. On the globality of motor suppression: Unexpected events and their influence on behavior and cognition. Neuron 93 (2), 259–280. http://dx.doi.org/10.1016/j.neuron.2016.12.013.

Wessel, J.R., Ullsperger, M., 2011. Selection of independent components representing event-related brain potentials: A data-driven approach for greater objectivity. NeuroImage 54 (3), 2105–2115. http://dx.doi.org/10.1016/j.neuroimage.2010.10.033.